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Utilizing Machine Learning to Advance Battery Materials Design: Challenges and Prospects

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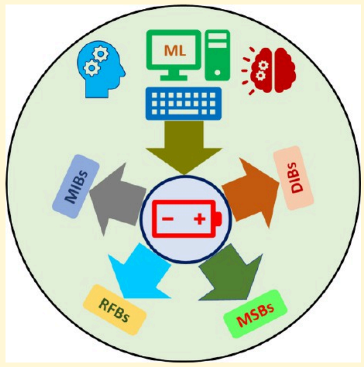
ABSTRACT: Advancement of batteries is indispensable for further utilization of renewable energy sources to meet the increasing energy demand. The rapid development of machine learning (ML) approaches has propelled innovation across diverse domains, fundamentally reshaping the landscape of energy storage research. This comprehensive and authoritative discussion critically examines the application of artificial intelligence (AI) and ML techniques for the design of materials for various battery systems by navigating a large material space. We emphasize recent progress in the battery field propelled by ML, describe existing and forthcoming hurdles, and elucidate the prerequisites for optimizing ML methodologies. We also provided future directions and potential research areas in the application of advanced ML techniques for the optimization of battery systems. The goal is to facilitate the transfer of these advanced AI/ML tools to researchers involved in battery design research, fostering a comprehensive understanding of their potential and embracing the multifaceted aspects of battery research.



1. INTRODUCTION

In the field of energy storage, batteries are the frontrunners starting from energy storage to transportation.[1] Batteries are highly intricate electrochemical systems consisting of a cathode, electrolyte, anode, current collector, and separator. Various types of batteries exist, including metal-ion, dual-ion, metal−sulfur, and redox flow batteries, each operating based on distinct mechanisms. Every battery class offers distinct advantages and drawbacks that are suited to particular industrial and everyday applications. Optimizing battery performance requires a thorough understanding of the interconnected relationships among the components and the inherent properties of the materials involved. Additionally, there is a crucial emphasis on optimizing devices for robustness and reproducibility. This optimization is vital for their integration into energy systems.[2]

Over the past decade, a plethora of experiments have been conducted alongside empirical and computational model-based simulation methods to advance our understanding of battery materials and their performance characteristics. In many instances, material selection for testing relies heavily on heuristic guessing informed by domain knowledge or inspired by previously reported materials. However, this approach, reliant on inefficient trial-and-error methods, involves testing materials one by one for the discovery of promising energy storage candidates. For battery development, selecting and synthesizing high-yield, high-quality materials from numerous possibilities for device integration typically takes over 15 years.[3,4] This process demands significant resources, including costly instrumentation and long-term research endeavors. Given the accelerating adoption of renewable energy sources, there is an urgent need for more rapid discovery of potential energy storage materials. Traditional methods are inadequate for achieving the necessary pace of discovery to meet these pressing demands.[5]

Data-driven material selection has become a trend for the acceleration of energy storage material discovery in the field of materials science. In recent years, the synergy between machine learning (ML) and big data has yielded unprecedented advancements across various disciplines.[6−9] While ML has traditionally thrived in computer and data sciences, its applications have expanded into physics, chemistry, biology, engineering, and materials science. Notably, ML technologies exhibit significant potential in the development and management of energy storage systems.[10−13] Using numerical representations of materials to predict their functions and optimize experimental reaction parameters, the integration of ML with traditional experimental and computational simulation techniques has significantly accelerated the material discovery process.[14−34]

Here, the discussion underscores the current trends in the ML-assisted investigation of potential energy storage materials for the design of a high-performance battery, focusing on specific aspects such as the utilization of various advanced ML

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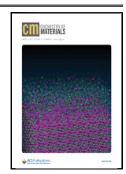

Scheme 1. Generalized Workflow of ML-Assisted Design of Screening of Materials for Batteries

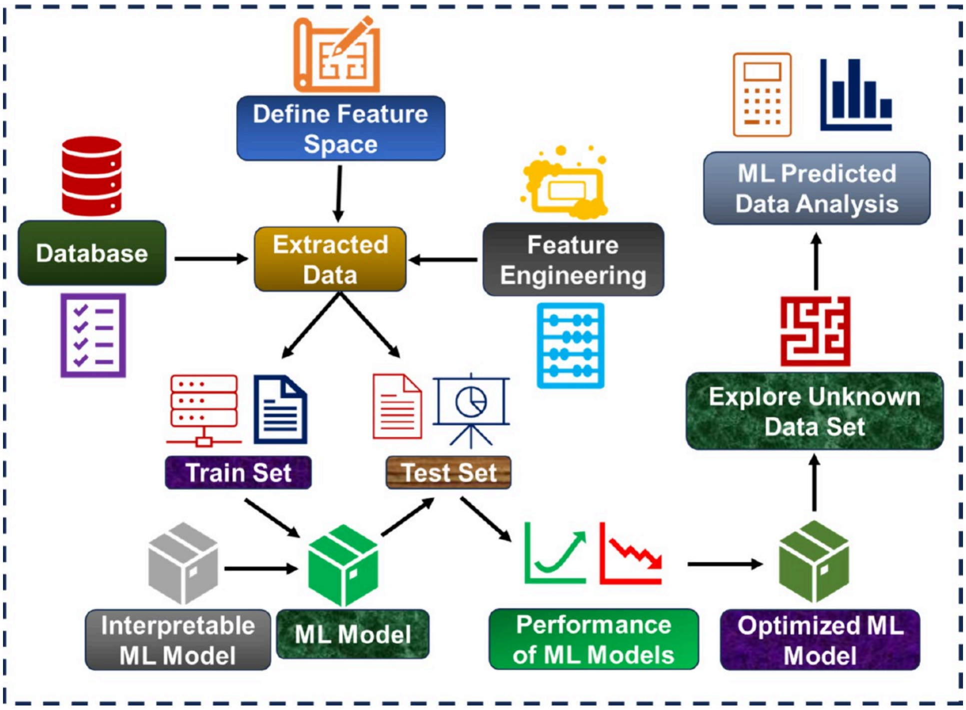

Table 1. Available Open Access Materials Databases and Their Descriptions

database	url	description
Materials Project	https://materialsproject.org	computational information on various class of materials, electrode materials, organic molecules
AFLOW	https://www.aflowlib.org	database of materials with a large number of properties
NOMAD	https://nomad-lab.eu/nomad-lab	repository for sharing the properties of materials
OMDB	https://omdb.mathub.io	electronic structure information on 3D-organic crystals
OQMD	https://oqmd.org	computed thermodynamics and structural properties of 1 226 781 materials
CCDC	https://ccdc.cam.ac.uk	crystal structure database
COD	https://www.crystallography.net/cod	crystal structures of inorganic, organic and metal−organic compounds from published works
NRELMatDB	https://materials.nrel.gov	materials of particular interest for renewable energy application

techniques. This Review essentially analyzes the recent progress in ML and artificial intelligence (AI) assisted design and optimization of various battery parameters, prospects, and unaddressed challenges. A generalized ML workflow (Scheme 1) is presented in the beginning of this Review, followed by a discussion on the availability of data and the challenges due to scarcity of data across the different type of battery systems. Furthermore, detailed discussions on the advancements in ML techniques—ranging from traditional models to neural networks, convolutional networks, and crystal graph neural networks—have been highlighted based on recent research trends, citing significant current developments. An elaborate and opinionated outlook given at the end details the ways in which further progress with ML and AI may occur and how it would guide experimental researchers in designing modern high-performance battery systems.

2. DATA AVAILABILITY

The fundamental building block for the development of any ML model is data. In the past decade, the development of databases dedicated to the storage of scientific data has become pivotal in advancing materials science, especially in the context of ML applications. These databases, such as Materials Project (MP), NOMAD, AFLOW, and OQMD, serve as repositories for a wealth of material properties. A list of open access databases, their URLs, and their descriptions is presented in Table 1.

There are two important parts of data used for any kind of supervised learning (regression or classification): the descriptor or target variable and features or input variables. The labeled data are needed for the training of the ML model, whereas the input variable is important for the efficient representation of the anode or cathode materials of the battery. While most material databases include general properties of materials, specific application-related descriptor data such as voltage, capacity, and energy density are often lacking. This lack of specialized data poses a challenge for developing accurate predictive models for battery applications. For instance, among the databases mentioned, only the MP database comprehensively covers 4351 electrode materials across 10 metal ions (Li, Na, K, Rb, Cs, Mg, Ca, Al, Zn, and Y) for batteries. It provides a rich data set of metal-ion battery properties including discharge voltage, specific capacity, energy density, and stability during charge−discharge cycles. However, due to the scarcity of dedicated databases, the application of ML is currently limited in certain areas, notably for batteries beyond metal-ion batteries such as dual-ion batteries, metal−sulfur batteries, and redox flow batteries. In addition to the labeled data, the amount of data also plays a key role in deciding the performance of ML models. A sufficient

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amount of data is necessary to develop a more generalized and robust ML model that can more accurately predict the battery properties.

A comprehensive database is crucial for the data-driven discovery of materials. While current data-mining efforts often rely on datasets generated through high-throughput experiments or theoretical simulations, an alternative approach involves constructing a database directly from scientific literature. A database of battery materials has been developed, comprising 292 313 data records that detail 214 617 unique chemical–property relationships across 17 354 chemicals and covering key material properties such as capacity, voltage, conductivity, Coulombic efficiency, and energy.[35] This database was autogenerated by mining text from 229 061 academic papers using an adapted version of the ChemDataExtractor toolkit specifically tailored for the domain of inorganic battery materials. The data extraction process involved leveraging APIs from publishers like Elsevier and the Royal Society of Chemistry to download over 229 000 papers, identified by the keyword "battery". A relevance flag ("R") was introduced to filter out less relevant records, ensuring data quality. The processed data, stripped of embedded markups and organized into plain text, were structured into a document object for further analysis. ChemDataExtractor, a specialized natural language processing (NLP) toolkit, was adapted to meet the specific needs of battery materials research. It employs advanced NLP techniques, including tokenization, part-of-speech tagging, and chemically named entity recognition (CNER). Key adaptations include the creation of regular expression rules for extracting chemical compounds from composite materials and the inclusion of suffixes relevant to nanomaterials and battery components, enhancing data classification. Central to the database's creation is relationship extraction, where specialized regular expressions tailored to the battery materials domain were used to identify relationships between chemical names, properties, values, and units. The database is publicly accessible, complete with a graphical user interface for ease of use, and serves as a valuable resource for accelerating innovation in battery technology. The Bidirectional Encoder Representations from Transformers (BERT) model has been adapted for battery research, leading to the development of six specialized models: BatteryBERT, BatteryOnlyBERT, and BatterySciBERT, each of which is comprised of both uncased and cased models.[36] These models were trained on a large corpus of battery-related research papers and fine-tuned for specific tasks, such as classifying battery papers and performing extractive question-answering to identify battery components such as anodes, cathodes, and electrolytes. The BatteryBERT models outperformed the original BERT models in these tasks, demonstrating their effectiveness in battery-related applications. Additionally, these models were employed to enhance a battery materials database, and a web application was developed for the interactive use and visualization. Zhang et al. introduced a comprehensive database for ionic conductors that is instrumental for advancing electrochemical devices such as batteries and fuel cells.[37] This resource contains transport properties for over 29 000 compounds, including cations (e.g., Li^+, Na^+, Mg^{2+}) and anions (e.g., F^-, O^{2-}). The database construction integrates Crystal structure Analysis by Voronoi Decomposition (CAVD) and Bond Valence Site Energy (BVSE) calculations, providing a dual approach to analyzing ion transport. CAVD identifies potential migration pathways by mapping interstitial sites and channel connectivity within crystal structures, with a focus on bottleneck dimensions relevant to ion migration. Complementarily, BVSE evaluates the energy landscape, calculating migration barriers and activation energies along transport pathways. This integration of geometric and energetic analyses uncovers migration paths optimized for conductivity. Validation against benchmark compounds further demonstrates the robustness of this CAVD-BVSE approach for screening fast-ion conductors. This data set is a valuable foundation for accelerated materials screening and the generation of ML descriptors, promoting large-scale research on ionic transport properties in inorganic compounds. Siqi Shi and co-workers introduced Solid Electrolyte Screening Platform for Solid Electrolytes (SPSE), a high-throughput tool that combines a materials database with automated ion-transport calculations.[38] The platform utilizes empirical algorithms, including geometric analysis and the bond valence site energy method, to preliminarily screen candidates and identify ion-transport pathways. These paths are then used to automatically generate images for precise FP-NEB calculations. With an open web interface, the SPSE database is accessible for ML integration. This platform enables efficient, automated screening, accelerating the discovery and design of advanced solid electrolytes. Recent advances in Generative Artificial Intelligence (GAI) illustrate its transformative potential in materials science, shifting from task-specific models to general-purpose applications. GAI technologies, supported by the prompt paradigm and RLHF, demonstrate promising capabilities in addressing complex tasks such as deriving structure–activity relationships across diverse materials. Notably, Liu et al. provide an in-depth analysis of GAI's applications, detailing methodologies, strengths, and limitations.[39] They underscore GAI's potential in tasks like inverse design of materials and data augmentation while acknowledging critical challenges and suggesting solutions, contributing to a roadmap for accelerating data-driven materials research. In many scientific fields, constructing high-accuracy models remains time- and labor-intensive. Auto-MatRegressor offers a promising solution by implementing metalearning to automate model selection and optimization, using metadata from historical data sets to refine future predictions.[40] With this approach, Auto-MatRegressor expedites the ML model-building process while maintaining high accuracy, making it a valuable tool for accelerating materials research. This advance aligns with the field's movement toward more automated, efficient, and scalable ML methodologies.

Apart from the target variable, representation of materials numerically through introducing essential features is a necessary step in the ML workflow. Overfitting is a fundamental challenge in ML applications, particularly in materials science, where data sets are often small, high-dimensional, and imbalanced. In such cases, the model memorizes training data rather than generalizing patterns to unseen data, leading to unreliable predictions. It is more common when the number of parameters in deep learning models significantly exceeds the number of training samples. A ML model with low cross-validation, training, and test errors, all in close proximity, is desirable for accurately predicting material properties in a more generalized manner. The efficacy of a ML model training is intrinsically linked to the judicious selection of appropriate input variables that can accurately capture the underlying patterns and relationships with the target variable.[17] Data sets are converted into numerical representations, often referred to as features. NCOR-FS is a feature selection method that integrates materials domain knowledge to improve ML predictions of materials properties.[41]

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By incorporating Non-Co-Occurrence Rules (NCORs), the method minimizes highly correlated redundant features, aligning feature subsets with domain knowledge. Using a swarm intelligence algorithm for optimization, NCOR-FS enhances the interpretability and accuracy of ML models. Compared to conventional feature selection methods, NCOR-FS achieves better feature subset quality and prediction performance, making it broadly applicable across materials systems. This approach exemplifies the value of embedding domain knowledge into data-driven algorithms for materials informatics. This process enables learning algorithms to discern meaningful patterns in the data. Dimensionality and the influence of features carry the applicability of ML models to learn effectively, operate efficiently, and provide interpretable and accurate results. Existing feature selection methods often require extensive hyperparameter tuning and do not consistently integrate domain expertise. To analyze the integrity behind the features and the target attributes, Siqi Shi and co-workers proposed a multilayer feature selection method DML-FSdek (Data-driven Multi-Layer Feature Selection incorporating Domain Expert Knowledge) to understand the perspective of different characteristics of data.[42] The DML-FSdek method is an innovative feature selection approach that integrates multilayer statistical evaluation with expert insights to refine feature selection for predicting the properties of materials. It operates through a three-layer structure, beginning with a sparsity layer that eliminates features with low variance or excessive sparsity, thereby discarding information that adds little to the model accuracy. Next, the correlation layer assesses the relationship between features and target properties using Mutual Information (MI) and the Pearson Correlation Coefficient (PCC), discarding features that fail to meet a minimum correlation threshold with the target. The redundancy layer further refines the feature set by removing highly correlated features, using the Distance Correlation Coefficient (DCC) and PCC to identify and eliminate redundant features that could lead to multicollinearity. Each layer's outputs are enhanced through the integration of domain expert knowledge, where a weighted scoring system incorporates expert-rated feature importance to retain key features, thus balancing algorithmic rigor with scientific relevance. The method dynamically optimizes thresholds within each layer to adapt to specific data sets, iteratively improving predictive power. Validated across diverse materials data sets, DML-FSdek achieves enhanced accuracy and interpretability compared to conventional methods like Lasso and Elastic Net. By combining multilayer statistical filtering with expert-guided adjustments, DML-FSdek not only accelerates the creation of accurate interpretable models but also sets a foundation for future improvements in automated feature selection with cumulative expert knowledge. The availability of limited data and high-dimensional feature spaces in materials informatics often hinders the performance of the ML models. The recent work by Liu et al. provides a comprehensive framework to address the challenges through a synergistic data quantity governance flow that incorporates materials domain knowledge.[43] The review highlights the importance of balancing sample size with feature dimensionality or model parameters and proposes strategies such as feature reduction, sample augmentation, and domain-knowledge-informed ML approaches to address the challenges. Currently, researchers are developing efficient text-mining framework for extracting valuable domain knowledge from the abstracts of published literature.[44] An automatic descriptor recognizer based on natural language processing (NLP) to identify latent descriptors for materials has been developed.[45] The method integrates a conditional data augmentation model with domain knowledge (cDA-DK) and employs coarse- and fine-grained descriptor recognizers (CGDR and FGDR). CGDR extracts descriptor entities from the literature with high accuracy (F1 = 0.87), while FGDR screens fine-grained descriptors for specific applications. Using NASICON-type solid electrolytes as a case study, 408 descriptors were identified, enabling the construction of predictive models for activation energy with excellent performance (R^2 = 0.96). This approach facilitates understanding of structure−activity relationships and accelerates materials discovery. Topological analysis and ML methods have been utilized to predict the glass transition temperatures of arsenic selenide (As_xSe_{1-x}) glasses, overcoming the limitations of costly and time-consuming experimental measurements.[46] Similar predictive methods can be applied in battery research to model temperature-dependent properties, aiding in the efficient design of advanced battery materials. A predictive model has been proposed for the glass transition temperatures of Ge_xSe_{1-x} glasses using feature selection-based two-stage support vector regression (FSTS-SVR).[47] Pearson correlation is used to select features, and SVR models are built separately for two structural stages and synthesized to minimize error. The model achieves high accuracy (RMSE = 10.64 K and MAPE = 2.38%) and outperforms other methods. However, when dealing with unstructured data, such as molecular graphs, inherent challenges arise due to the presence of invariable characteristics that are difficult to capture in a vectorial representation.[48] Numerous strategies have been proposed to address this challenge, including invariant integration, parameter sharing, density representations, and fingerprinting techniques.[48] One avenue of exploration involves the utilization of neural network (NN) models to infer representations directly from data.

3. MACHINE LEARNING IN BATTERIES

3.1. Metal-Ion Battery. A metal-ion battery works by shuttling metal ions (like lithium) between two electrodes, the anode and the cathode, during charge and discharge. When charging, an external power source moves ions from the cathode to the anode via an electrolyte, while electrons flow through an external circuit to balance the charge. During discharge, ions migrate back to the cathode and electrons flow in the reverse direction to power a device. Currently, secondary rechargeable metal-ion batteries (MIBs) are the most in-demand and the most explored area in the energy storage field. In the following sections, we demonstrate the utilization of ML models for the prediction of various battery properties and their advantages and disadvantages, followed by the identification of problems that have not yet been solved and their possible solutions.

3.1.1. Electrode Materials. 3.1.1.1. Voltage. The research in this field has progressed a lot, covering both experimental and computational perspectives. Due to extensive research on metal-ion batteries, a lot of experimental and computational data are available for the training and testing of ML models, which is an extra edge for MIB research over other types of batteries. In addition, there are databases (Table 1) such as the Materials Project database that contain a large amount of different metal-ion battery data, which have been utilized by various research group to develop ML models for the prediction of various battery properties.

Barone and co-workers used deep neural network (DNN), support vector machine (SVM), and kernel ridge regression

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(KRR) for the prediction of voltage based on the elemental descriptors considering electrode materials from the MP database.[49] 237 elemental features have been considered to represent an electrode material, including the working ion. Principal component analysis (PCA) has been adopted to reduce the feature space to 80 features. The DNN model is composed of four layers, with 80 input nodes, 60 first hidden layer nodes, and 30 second hidden layer nodes, followed by one output node for the prediction of voltage by considering mean-square-error as the loss function, MAE as the performance metric, and "rmsprop" as the optimizer. The parameter tuning has been done on a 90% train data set, and the performance of ML models has been evaluated using a 10% hold data set. The cross-validated train errors for DNN, SVR, and KRR are 0.47, 0.50, and 0.51 V, respectively, whereas the test errors are 0.42, 0.44, and 0.44 V. The close proximity between the cross-validated train and test errors indicates the absence of any overfitting issue. In addition, the ML models are also trained on only Li data for the prediction of Na and K data, showing comparatively large error with a MAE of 0.62−0.70 V. Though the models are quite generalized, the reduction of feature space using PCA may lead the loss of some important chemical information that might help to increase the accuracy of the ML models. Consideration of more efficient features or graphical representation of electrode materials may increase the accuracy of ML models, as the elemental features would not be able to distinguish between two same chemical formula having different structures as well as the connectivity among the atoms in a crystal structure. Also, the division of whole data set into 90% train and 10% test sets needs to be justified. It has been observed that none of the ML models were able to capture the low-voltage region, which may be because of the limited amount of data present in the low-voltage range in the training set; the addition of further voltage values in that range might mitigate the problem. Furthermore, it would be beneficial to provide a clear and detailed description of the parameter tuning process for the DNN architecture. The code used for such studies should ideally be made available in a public repository so that other researchers can replicate and build upon the methodology.

In their next work, Barone et al. showed that the proposed DNN model can be utilized for the prediction of voltage along with the volume change of electrode materials with moderately high accuracy (MAE of 0.41 V) considering compositional features generated from the matminer package.[50] The matminer package is an efficient tool for generating various kinds of features in a very short time. The reported mean cross-validation error and test errors for all three utilized data sets (All, Alkali, and Li) are in the range of 0.37−0.43 V, which indicates no overfitting in the ML-predicted voltage value. Twenty electrode materials that have been explored for Li ion batteries but not investigated for Na-ion batteries were proposed by the DNN model. The DNN model was found to outperform the traditional ML models such as KRR and SVM. However, interpretation of such a complex DNN model is much more difficult compared to traditional ML models.

Though ML models are trained on voltage data of different metal-ion batteries, the amount of data is still too small (~4351 in MP) to leverage the high potential of various advanced neural network models. However, there is a large number of materials present (>130 000 in MP) whose formation energies are reported. Thus, the average voltage of a battery can be related to the energy difference between the reactants and products of the intercalation reaction at the two electrodes. When a metal ion (e.g., Li^+, Na^+) intercalates into or deintercalates from the host electrode, a redox reaction occurs, leading to a change in the Gibbs free energy of the system. The average voltage of this reaction can be derived from the Gibbs free energy change per ion transferred. For an intercalation reaction, the average voltage, V, can be expressed as

$$V = -\frac{\Delta G}{nF}$$

where ΔG is the change in Gibbs free energy per formula unit, n is the number of electrons transferred, and F is Faraday's constant. ΔG can be approximated to by the change in formation energy (ΔE) because the entropic contributions are relatively small in solids compared to liquids or gases. This leads to the following approximation:

$$V \approx -\frac{\Delta E}{nF}$$

where ΔE represents the formation energy difference between the lithiated and delithiated states of the material. For example, if we calculated formation energies E_{LiMO_2} and E_{MO_2} for lithiated and delithiated cathode materials, respectively, the change in formation energy can be expressed as

$$\Delta E = E_{LiMO_2} - E_{MO_2} - E_{Li}$$

where E_{Li} is the energy of the metallic Li bulk per atom. Thus, by calculating or predicting the formation energy of the intercalated and nonintercalated states, one can estimate the average voltage of the battery. As we know, the amount of data is a crucial factor for training any ML models, and it is always a good practice to train a ML model with a larger data set, which must be more accurate; from that accurate prediction, one can easily calculate the average voltage of a battery. Moreover, there are already well-established ML models trained on reported formation energy data, which can be utilized for the prediction of the formation energy for unknown material space (https://matbench.materialsproject.org/Leaderboards%20Per-Task/matbench_v0.1_matbench_mp_e_form/). In addition, transfer learning (TL) can significantly enhance voltage prediction in batteries by building on models trained on formation energy data. Since formation energy and battery voltage are intrinsically connected through the thermodynamics of intercalation reactions, a model trained on formation energy already possesses critical insights into how structural and compositional features affect material stability. This underlying knowledge is crucial for voltage prediction, where the model must understand how chemical properties and energy states shift when ions move between electrodes.

In practice, this approach begins with training a model on a large formation energy data set, such as those available through resources like Matbench or the Materials Project. During this training, the model captures fundamental patterns related to material composition, atomic structure, and bonding, effectively learning which chemical features contribute to energy stability. Once trained, this model has a strong foundation in core material science concepts that are closely related to, though not identical with, the requirements for voltage prediction. By transferring this model to a new task focused on voltage, one can retain the layers responsible for these chemical insights while refining the model further for voltage-specific data. This process is particularly beneficial, since voltage data are typically more limited than formation energy data. TL enables effective use of

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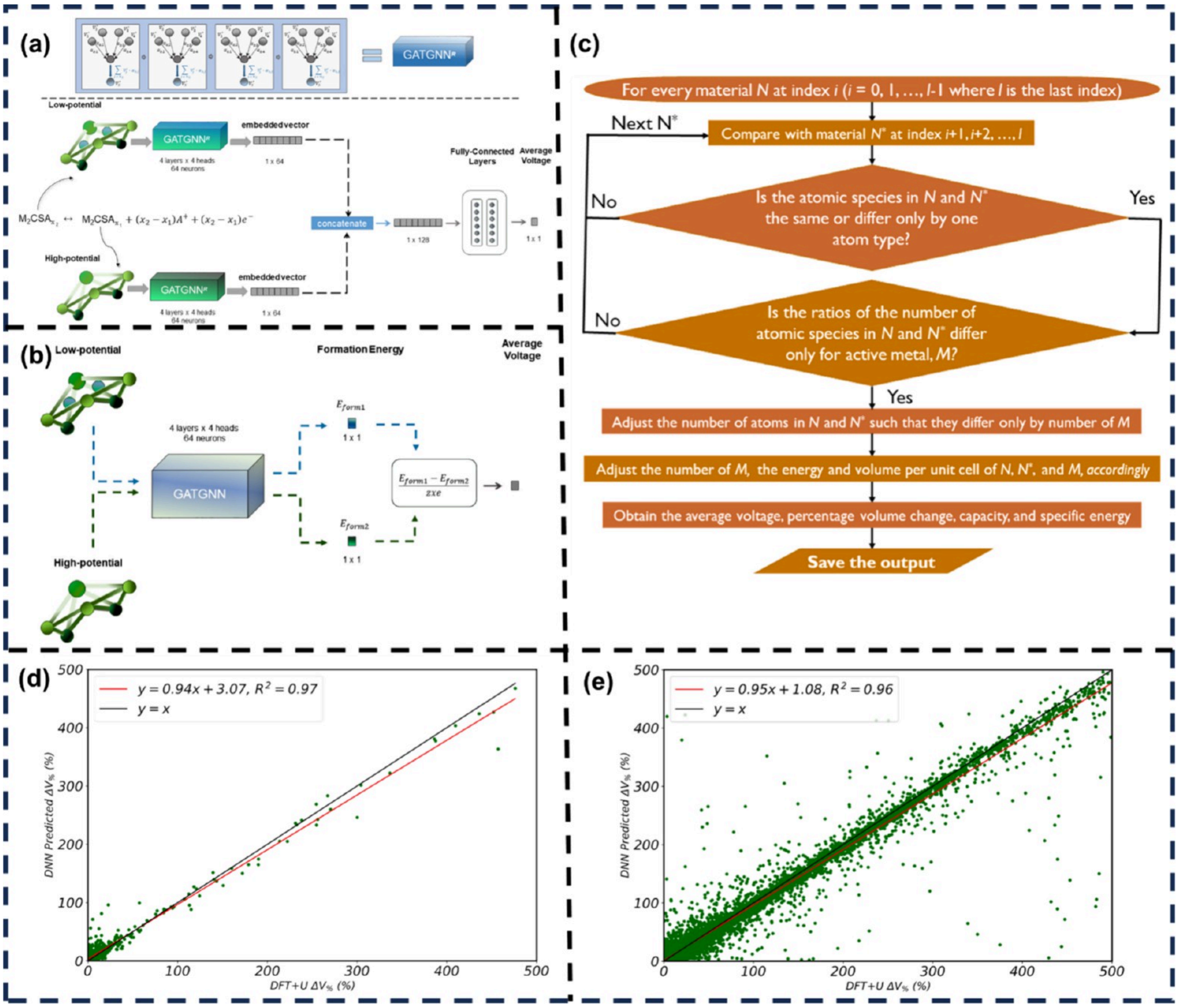

Figure 1. ATGNN workflow for the learning of average voltage of electrode materials for two different methods: (a) prediction of voltage directly from the reaction between the electrode material and intercalated ion and (b) determination of average voltage from the ATGNN-predicted formation energy. Reproduced from ref 52. Copyright 2022 American Chemical Society. (c) Data mining workflow to navigate a materials database (MP and AFLOW) for the expansion of the battery database and (d, e) parity plots for the comparison of the volume change of electrode materials for the intercalation and deintercalation of metal ions upon charging and discharging for two different data sets. Reproduced with permission ref 53. Copyright 2022 Elsevier.

this sparse data while minimizing the risk of overfitting by relying on the foundational knowledge from formation energy patterns.[51] Further, Hu and co-workers have improved the accuracy in the prediction of voltage by using an attention-based graph neural network (ATGNN) based on the material's spatial information and compositional features from the constituent atoms in the 3D space.[52] The ATGNN is a type of neural network designed for processing data structured as graphs. It incorporates attention mechanisms, which allow the model to focus on specific parts of the graph while making predictions. The graph representation includes both the atom and bond features of the materials. The graph representation of materials is a more advanced representation compared with only consideration of the elemental and compositional features. Two different models have been proposed for the determination of the average voltage. The first model is able to learn the average voltage directly from the reaction between electrode materials and the working ion, and the other model learns the formation energy of the material, from which voltage can be calculated. The authors used two different data sets, one of them being the data set from the battery app of MP for the training of the first ML model and the other 60000 materials collected from the materials explorer app of the MP database along with the formation energy, and proposed a relationship between formation energy and voltage. Therefore, based on the chosen methods, the target variable voltage and the hold-out test MAE (0.344 and 0.315 V) decreased for both data sets compared to the previous report.[52] The ATGNN ML workflow for the prediction of the average voltage from the reaction mechanism and from the formation energy is depicted in Figure 1a and b. Extraction of data in efficient way is also important, though it can be time-consuming if done manually. Despite the demonstrated advantages of ATGNNs in capturing intricate spatial and compositional features of materials, several challenges remain. Notably, ATGNNs exhibit increased computational complexity and sensitivity to hyperparameter tuning, which can limit their scalability and robustness. Additionally, the model's capacity to focus on relevant graph components through the attention mechanism may inadvertently lead to overfitting, especially in scenarios with limited training data. Regularization techniques and careful cross-validation become essential to overcome the overfitting issue.

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Table 2. Application of Various ML Algorithms for the Prediction of Material Property (Target Variable) and the Major Outcomes

ML algorithms	target variable	major outcome
KRR, SVR, DNN	average voltage	trained the ML models with ∼5000 electrode materials based on the elemental features to propose potential cathode materials for Na and K-ion batteries[49]
ANN, RF, GBM, SVM, KRR, KNN	discharge capacity of first and 50th cycle	GBM model successfully predicted the discharge capacities of NCM oxide cathode materials based on the structural and elemental features; higher Li concentration with low concentration of electronegative dopant atom led to higher discharge capacity[55]
RR, LR, DNN, DT, RF, GBM, SVM	discharge capacity of first and 20th cycle	GBM model predicted specific capacity with higher accuracy based on the fundamental material properties such as molar mass, dimension of crystal lattice[56]
DNN	average voltage, volume change	predicted electrode materials for Na-ion battery by replacing Li by Na from parent host material based on the higher average voltage and small volume change[50]
ATGCNN	average voltage	significant improvement in average voltage; proposed two models based on the composition and arrangement of atoms in lattice crystal to learn voltage and formation energy, successfully reported 10 different fluorophosphate battery materials with average voltages >3.1 V for a Na-ion battery[52]
DNN	data mining, average voltage, volume change	provided efficient data mining flowchart for selection of charge and discharge state of various metal-ion batteries, navigated MP and AFLOW databases to propose cathode material having high DNN predicted average voltage[53]

Moreover, preparing an unknown data set to explore innovative electrode materials requires an advanced methodology for selecting the right materials that belong to the training and testing space. Recently, Moses et al. developed an efficient data-mining protocol to navigate all the inorganic crystal materials from the MP and AFLOW databases (Figure 1c).[53] They have extracted materials with both possible charge formulas and discharge formulas including the working ion. With this data-mining protocol, they expanded the battery database from ∼5000 to 190 000 instances. Further, a regression-based DNN has been built on top of the expanded battery database for the prediction of average voltage along with the volume change of the electrode material upon charging and discharging. The parity plot (Figure 1d and e) shows the stability of the DNN model and accurate prediction of volume change through the comparison between DFT-calculated and DNN-predicted volume change. The trained DNN model using an expanded database demonstrated a 28% improvement in accuracy compared to previously reported results for predicting voltage and volume change. This underscores the critical role of high-quality data volume in training ML models, particularly for DNNs.[53] This model successfully predicts properties for both the anode (negative voltage region) and the cathode (positive voltage region), addressing a major drawback of previously reported ML models.

Though all the ML models have been able to predict the specific battery properties accurately within the considered data set, lack of chemical insights is a drawback of these studies. Along with regression, application of an unsupervised clustering method could cluster down the materials with similar characteristics, which can shed light on which kind of materials shows which range of target properties like voltage or capacity.

3.1.1.2. Specific Capacity. Specific capacity is also an important parameter in determining the performance of battery materials. Void-identification techniques can be used for estimating the theoretical maximum specific capacity by identifying the number of electroactive ions that can intercalate within a structure. However, this method alone may not suffice for accurately predicting specific capacity across different working ions and electrode compositions. The size and properties of working ions (e.g., Li^+, Na^+, K^+, Zn^{2+}, Ca^{2+}, Al^{3+}) vary significantly, influencing their ability to intercalate into host structures depending on factors like charge, ionic radii, and charge/mass ratio, among others. Ion intercalation energetics and structural compatibility differ between ions, resulting in loading and thus varying specific capacities. This is where ML models trained on DFT data can offer significant advantages in screening large material data sets. In a recent study, Wang et al. showed the value of integrating structural, chemical, and physical properties in ML models.[54] By using a modified crystal graph convolutional neural network (mCGCNN), they achieved a lower mean absolute error (MAE) compared with traditional methods by incorporating these features. This highlights the importance of combining domain knowledge with advanced computational tools for more accurate specific capacity predictions. Wang et. al predicted the discharge capacity of the initial and 50th cycle for the nickel−cobalt−manganese (NCM) oxides cathode materials.[55] They used 6 nonlinear ML algorithms, among which gradient-boosting model (GBM) has been found to be suitable for the learning of discharge capacity. Small RMSEs of 16.66 and 18.59 $mAhg^{-1}$ for the initial and 50th cycle discharge capacity prediction, respectively, have shown the potential of the boosting algorithm to learn the underlying structure−property relationship for the NCM cathode materials. They also carried out a game-theory-based analysis and concluded high Li concentration with a low concentration of the electronegative dopant atom can possess high discharge capacity. However, several concerns need to be considered. First, the reliance on complex deep learning architectures such as mCGCNN raises the possibility of overfitting, especially if the training data set does not adequately represent the full variability of NCM cathode materials. Although the reported RMSE values are promising, the absence of a thorough uncertainty quantification or validation on independent data sets may limit confidence in the model's predictive robustness under different conditions. Moreover, while the use of multiple nonlinear ML algorithms offers a broad perspective, the trade-offs among predictive accuracy, interpretability, and computational cost remain insufficiently addressed. Finally, the game theory-based analysis, which links high discharge capacity to high Li concentration and low concentrations of electronegative dopant atoms, provides an interesting heuristic but may oversimplify the complex interplay of factors governing electrode performance. This suggests that further experimental validation is needed to substantiate these correlative insights and to confirm their causal significance in practical applications.

A similar kind of material screening was carried out by Wang et al. for a spinel class of materials, where they predicted the discharge capacity of the initial and 20th cycle with RMSEs of

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11.90 and 11.77 mAhg^{-1}, respectively, using the GBM model.[56] The accurate prediction of discharge capacity underscores the excellent predictive potential of the bagging-boosting based algorithm. Since the ML model is trained on the experimental capacity data for a specific class of materials, the generalizability of the GBM model needs to be further examined by exploring materials beyond training and testing data. Noteworthy utilization of various ML models to identify potential electrode materials for rechargeable batteries is presented in Table 2.

In recent years, graph neural networks (GNNs) have garnered significant attention in materials science for their ability to accurately represent complex material structures by capturing atomic environments. A graph, composed of nodes (representing atoms) and edges (representing bonds), can effectively model both molecular and crystalline structures. The Crystal Graph Convolutional Neural Network (CGCNN), developed by Grossman and colleagues, stands out as a pioneering model in this domain.[57] It employs a nearest-neighbor algorithm to iteratively update node features, thereby enhancing the representation of the local chemical environments. This model has inspired various adaptations, yet its application has primarily been confined to general material property prediction, with limited exploration in battery-specific contexts. One of the key challenges in applying GNN models, such as CGCNN, to battery research, especially beyond lithium-ion systems, is the scarcity of large high-quality data sets necessary for training. This limitation of data is a significant bottleneck in the field. Transfer learning emerges as a promising solution to this problem, where a pretrained model, originally trained on a large data set from a different but related domain, can be fine-tuned for a new task with minimal data. Recent advancements have demonstrated the potential of integrating transfer learning with CGCNN to predict the voltage of electrode materials for post-lithium-ion batteries using limited data.[51] The CGCNN model, initially trained on a substantial data set of lithium-based materials, was fine-tuned through transfer learning to predict the voltages of other metal-ion batteries. This approach not only overcame the data scarcity issue but also outperformed traditional machine learning algorithms in terms of accuracy. The success of such advanced techniques underscores the need to expand their application in battery research, enabling more precise predictions of battery properties and facilitating the design of efficient electrode materials.

3.1.1.3. Electrolytes. Apart from electrode materials, the electrolyte also plays a crucial role in determining the stability, charging rate, and voltage in batteries. There are not many reports on the ML-assisted optimization of electrolytes for batteries. Most of the investigation on solvent electrolytes has been done based on DFT study. Selection of the suitable solvent electrolyte can improve the performance of a battery. ML models can be utilized to find a suitable electrolyte for rechargeable batteries by exploring a large number of solvent molecules.

Ishikawa et al. utilized different ML models for the prediction of coordination energies of five different alkali metals (Li, Na, K, Rb, Cs) with 70 different solvent molecules.[58] They utilized easily available experimental and computational properties of solvent molecules as inputs for the training of ML models. Three ML models, namely, multiple linear regression (MLR), LASSO, and exhaustive search with linear regression (ES-LiR), were utilized to establish a relationship among the solvent molecules and the coordination energy. The authors showed an exhaustive search with Gaussian process (ES-GP), where the Gaussian-type regression method was able to improve the prediction accuracy of the coordination energy compared to the ES-LiR model with a very small number of data points.[58] Consideration of the HOMO energy and LUMO energy as features may not be a better choice, as exploration of large solvent electrolyte space will then require the calculation of the HOMO−LUMO energy of each solvent electrolyte. Also, the consideration of configuration for the solvent−metal coordination is very simple, where the metal is coordinated with only one solvent molecule and is not able to capture the real scenario. Further studies incorporating cointercalation phenomena can help to a more comprehensive analysis. The study indicates that Gaussian-based algorithms can be more effective with a very small number of data points, whereas ensemble-boosting based ML models can efficiently learn the interaction energy from simple elemental and structural features of solvent electrolytes. The compatibility between solvent electrolytes and high-voltage electrode materials remains a critical challenge in the advancement of rechargeable metal-ion batteries. Improving the prediction of electrochemical windows (ECWs) for battery electrolytes is a critical factor for pairing with high-voltage cathodes to enhance the energy density. Traditional methods based on single-electronic approximations, such as HOMO/LUMO calculations, show significant deviations (mean absolute errors up to 3.25 V) from experimental ECWs due to their inability to account for complex solvent interactions. To address this, a thermodynamic cycle-based approach has been developed that incorporates reorganization energy and solvation energy corrections, which are quantified using two geometric descriptors (λ and ΔG_{sol}), reducing the error to below 0.68 V.[59] Using this method, researchers developed a database of 308 electrolyte solvents and identified two high-performing solvents with ECWs exceeding 6.00 V and excellent structural stability. This work highlights the importance of accurate ECW prediction and descriptor-based screening for the rapid discovery of advanced battery electrolytes. The rapid identification of suitable solvent electrolytes with optimal electrochemical windows (ECWs) within the vast chemical search space is a formidable task, demanding innovative approaches to overcoming this bottleneck. Recently, we introduced an innovative approach to overcome the challenge of identifying compatible solvent electrolytes for high-voltage electrode materials in rechargeable metal-ion batteries.[60] Through the integration of supervised and unsupervised ML techniques, a vast pool of 4882 solvents was rapidly screened. The supervised ML model achieved high accuracy in predicting solvents with optimal ECWs, outperforming traditional DFT methods. A two-step feature selection process such as Select-K-Best and model-dependent feature importance helped to reduce the feature space by eliminating fewer contributing features, which finally resulted in the development of a robust ML model. Additionally, the use of K means clustering (unsupervised ML) efficiently categorized the solvents into 11 distinct clusters based on their ECW ranges. This dual approach not only accelerates the identification process but also reduces the solvent search space, significantly streamlining experimental efforts in battery research. Wang et al. explored the crucial role of ESWs in assessing the compatibility of solid-state electrolytes (SSEs) and coatings with electrode materials in metal-ion batteries.[61] The authors present an advanced prediction method that dynamically determines direct or indirect decomposition pathways, showing a good correlation with experimental oxidation and reduction potentials. Using this method, they established a

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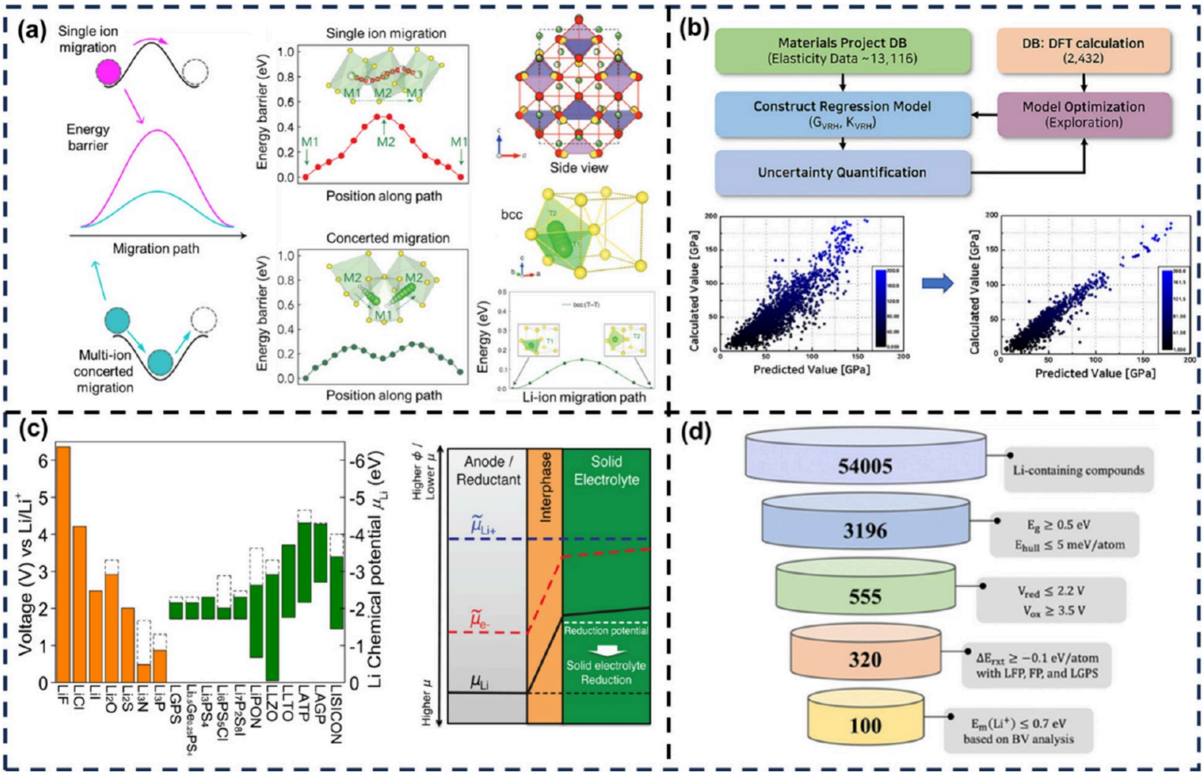

Figure 2. (a) The diffusion mechanism of migratory ions between different sites and different anion frameworks for solid electrolytes. Reproduced with permission from ref 69. Copyright 2018 Cell Press. (b) The flowchart and working principle for ML to accelerate the search for potential Na-solid electrolytes with desirable mechanical properties at minimum cost. Reproduced from ref 71. Copyright 2021 American Chemical Society. (c) Electrochemical window of solid electrolytes and coating materials. Reproduced from ref 72. Copyright 2015 American Chemical Society. (d) Screening of promising coating materials for a lithium−iron−phosphate cathode material. Reproduced with permission from ref 73 Copyright 2024 Royal Society of Chemistry.

reproducible database containing phase diagrams and electrochemical stability data for over 1500 solid-state inorganic compounds, with migrating ions including Li^+, Na^+, K^+, Mg^{2+}, Ca^{2+}, and Al^{3+}. This growing database offers valuable insights into the structure−activity relationships for the electrochemical stability of these materials and demonstrates the potential of ML to accelerate the screening of superior SSEs and coatings. In the pursuit of advancing battery technologies for transportation and aviation, the integration of robotics and machine learning has emerged as a powerful tool to accelerate electrolyte optimization. A innovative approach combining a custom-built automated experiment system, "Clio", with a Bayesian optimization-based experiment planner, "Dragonfly", has demonstrated a 6-fold acceleration in identifying fast-charging nonaqueous electrolyte solutions for Li-ion batteries.[62] Over just 2 days and 42 experiments, six high-performing electrolyte formulations were discovered, far outpacing traditional random search methods. This accelerated optimization process was further validated by improved fast-charging performance in practical graphite||$LiNi_{0.5}Mn_{0.3}Co_{0.2}O_2$ pouch cells, highlighting the potential of this integrated approach to rapidly advance battery innovations that are crucial for electrification.

3.1.2. Solid Electrolytes. Alternative to flammable organic electrolytes, solid-state electrolytes (SEs) with high ionic conductivity have shown promising results in terms of safety, long cycle life, and higher stability.[63−66] A quite popular lithium superionic conductor electrolyte, $Li_{10}GeP_2S_{12}$ (LGPS), has shown an extremely high ionic conductivity of 12 mS/cm at room temperature that is comparable to or higher than those of many other organic liquid electrolytes currently used in lithium-ion battery systems.[67] The development of alkali (lithium and sodium) superionic conductor electrolytes (SICEs) has mainly relied on alkali ion conduction. Some of the studied SICEs are mainly natrium superionic conductor (NASICON), lithium superionic conductor (LISICON), perovskite, antiperovskite, garnet, and thiophosphate structures.[68] Along with the conductivity of these SEs, their stability and compatibility with the electrode materials were studied rigorously. The diffusion mechanism of the migratory ion in between different sites and a different anion framework is shown in Figure 2a.[69] Researchers have used different simulation techniques such as first-principles simulation, ab initio molecular dynamics (AIMD) simulation, and Monte Carlo simulation for designing SEs.[70] AIMD simulations can also be used to quantify detailed diffusional information, such as site occupancy, concerted migration, jumping rate, correlation functions, and radial distribution functions, which are crucial for understanding migratory ion diffusion through grain boundaries and also in the bulk crystal.[70]

Despite their potential utilization, SEs face several challenges in achieving optimal performance and compatibility with the electrode materials. The searching space is too high to apply a simulation method for the design of a large number of SEs. By leveraging ML algorithms, researchers can rapidly screen a large number of potential candidate materials for SEs by predicting multiple properties such as ionic conductivity, electrochemical window, bulk modulus, and so on. Performing material substitution followed by the ML-assisted design of hypothetical SEs has become popular in the past few years.[74] ML-based

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approaches can accelerate the selection of superionic conductor materials up to thousands of times faster than the guess and check research pattern.

Logistic regression, a classification algorithm has been used to identify 21 potential SEs from a pool of 12 000+ Li-ion-containing materials that are fast ion conductors.[75] Min and co-workers have adopted a data-driven approach for the search of mechanically superior SEs for lithium-ion batteries from the data sets of 17 619 candidates.[76] They successfully reported potential SEs using the LightGBM model trained on top of chemical and structural features for the accurate prediction of elastic properties, which were further validated by the first-principles calculations. While the LightGBM model demonstrates promising predictive capabilities, its reliance on predefined chemical and structural features might overlook key descriptors influencing the elastic properties. Additionally, validating only via first-principles calculations may not fully capture real-world experimental complexities, warranting further experimental corroboration. The authors also showed the ultrafast screening potential of ML models for the selection of suitable SEs from a pool of more than 20 000 materials.[77] Nine classification algorithms (Naïve Bayes (NB), Logistic Regression, DT, Stochastic Gradient Descent, KNN, RF, Gradient Boosting (GB), Light Gradient Boosting Machine (LGBM), and eXtreme Gradient Boosting (XGB)) have been tested to classify the materials as superionic or nonsuperionic conductors. Among the applied ML models, the RF and LGBM models were found to perform exceptionally well with very high accuracy values of 0.887 and 0.886, respectively. First-principles calculations were also performed on the validation set to justify the model's accuracy. As a result, the authors reported lithium-solid electrolytes (Li-SEs) that have not been previously studied or investigated. The screening approach shows promise in rapidly narrowing down candidate Li-SEs; however, potential biases in the training data and overfitting may limit the model's generalizability. A Hierarchically Encoding Crystal Structure (HECS) method has been implemented to identify descriptors for Li^+ conduction in SSEs.[78] Using cubic Li-argyrodites as a case study, 32 HECS-derived descriptors were constructed, capturing the composition, structure, and conduction pathways. Partial correlation analysis revealed that smaller anion size is key to lowering the activation energy (E_a), which is influenced by lattice space and bottleneck size. Promising candidates, such as $Li_{5.5}PS_{4.5}Cl_{1.5}$ (ionic conductivity: 9.4 mS/cm, E_a = 0.29 eV), were experimentally validated, showcasing the excellent performance. This work provides a systematic approach to correlate descriptors with Li^+ conduction, which is applicable to other SSEs. The partial least-squares (PLS) technique reduces independent variables to a smaller set of uncorrelated components, then performs least-squares regression on these components. This method is particularly effective when multicollinearity exists among the independent variables. The HECS method effectively captures a range of descriptors, yet its application to only cubic Li-argyrodites raises questions about its broader generalizability to other SSE systems. Moreover, relying on partial correlation analysis and PLS for reducing multicollinearity may oversimplify the complex interplay of factors influencing Li^+ conduction, suggesting that further validation across diverse materials is needed. The PLS algorithm has been used based on HECS descriptors to predict and optimize the E_a for Li^+ conduction in SSEs, specifically cubic Li-argyrodites with a same train and test RMSEs of 0.2 eV.[79] The training and testing RMSEs show that the model is a generalized model without any overfitting and can be utilized for the prediction of E_a for unknown SSEs. Achieving high prediction accuracy, the model identifies key structural factors such as anion size and site disorder that influence E_a, thereby enabling the design of further Li-argyrodites SSEs with lower E_a. Though many classification algorithms have been used to classify a SE as conductive or nonconductive, studies on the utilization of regression algorithms for the prediction of SE's property are very limited. Tuning multiple properties simultaneously is necessary for the design of suitable SEs for rechargeable batteries.

The design of SEs lies in the choice of primary components and then the construction of soft frameworks so that the ions can move quickly by following a minimum energy pathway. Neural network-based first-principles calculations can speed up the search for structure prediction and mechanistic pathway investigations by 3−4 orders of magnitude without significantly sacrificing accuracy.[80] The recently developed Stochastic Surface Walking global optimization combined with global Neural Network potential (SSW-NN) is a very powerful and efficient method for the prediction of the crystal structure and transition pathway.[81] The Feedforward neural network model, which has used input features like group number, period number, and coordination number for the prediction of ionic conductivity in a SE material, has come under consideration for material selection in very short span of time. While these neural network-based methods greatly accelerate structure prediction and pathway analysis, their reliance on simplified descriptors may not fully capture the complexity of ion conduction mechanisms. Moreover, potential overfitting and generalizability issues call for further experimental validation. Kim and co-workers have reported models that can be trained using the Li diffusivity data of ternary compounds and can accurately predict the same for quaternary compounds.[82] This approach helps them tackle complex simulations more effectively. They have applied AIMD and interatomic ML potentials using the Sparse Gaussian Process Regression (SGPR) algorithm for the modeling of Li diffusivity in various SE composites.[82] It is well understood that mechanical properties keep the interface stable. So, to accelerate the search for Na-SEs with preferential mechanical properties at minimum cost, development of an ML regression model that can accurately predict the mechanical properties is very important. Min and co-workers have developed a LightGBM model for the prediction of shear and bulk moduli with a MAEs of 11.8 and 15.3 GPa, respectively.[71] The working flowchart and ML-predicted mechanical properties are illustrated in Figure 2b. The authors considered more than 12 000 candidates from the MP database with their chemical and structural descriptors. The trained ML model predicted the mechanical properties of ∼2000 materials, which was further validated by DFT calculation.

The interphase engineering between the electrode and the SE plays a very important role in electrochemical performance. The premise of artificial coating layers is to stabilize the interface and reduce the interfacial resistance. The electrochemical window of the coating material plays an important role in the chemical stability of the battery. Appropriate selection of the coating material and its compatibility with the neighboring composites impart stability to the battery (Figure 2c).[72] To mitigate the interface issue, introduction of a buffer layer/coating between the SE and electrode material is a strategy to inhibit the contact. The experimental approach for finding the perfect coating material is a time-consuming process, while computational approaches are quite fast and efficient. Shi et al. performed high-

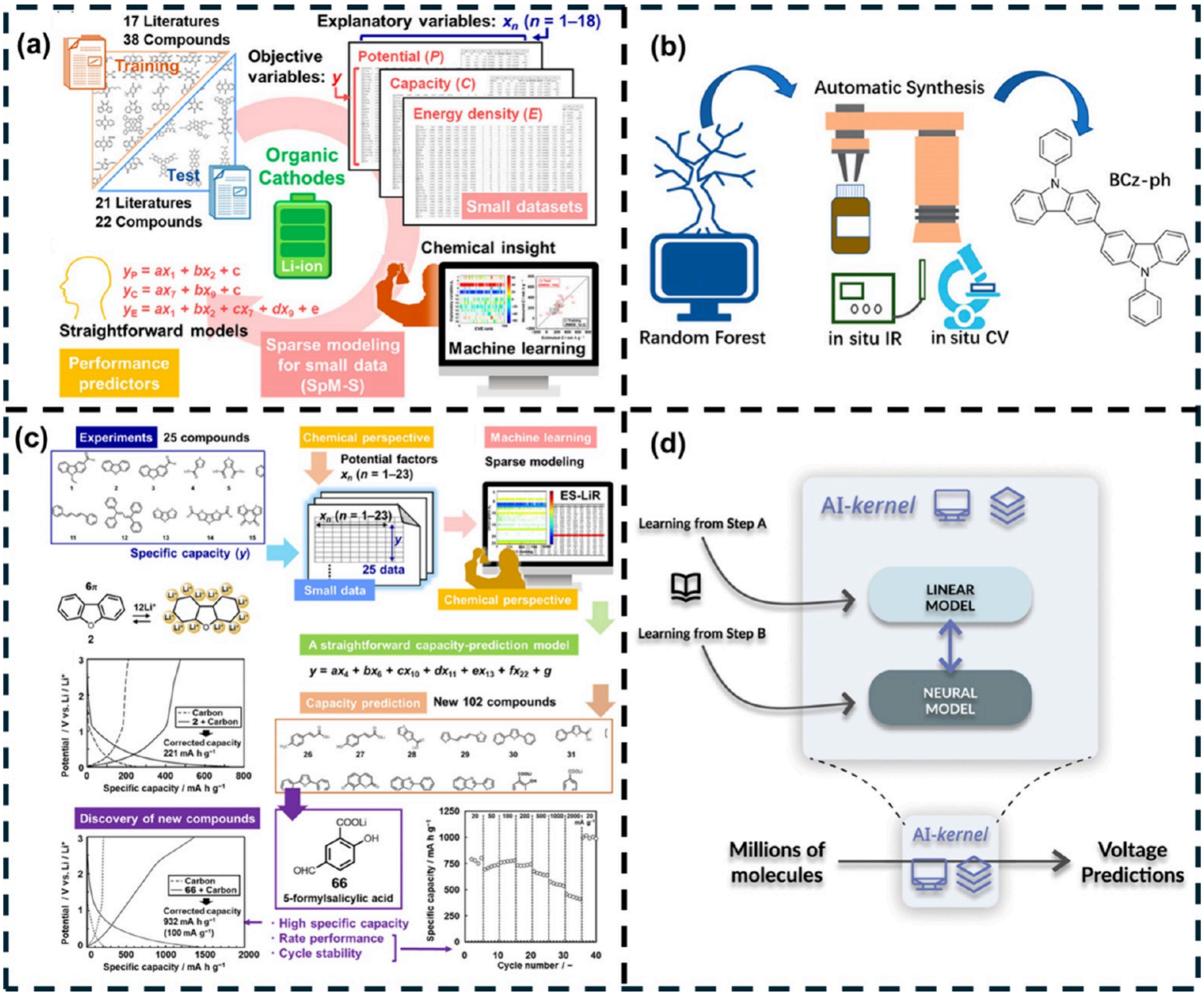

Figure 3. ML-assisted screening of potential organic electrode materials. (a) ML workflow to construct the model of performance predictors for a small data set. Reproduced from ref 90. Copyright 2022 American Chemical Society. (b) Utilization of an interpretable Random Forest model for the design of a high-voltage organic electrode. Reproduced from ref 91. Copyright 2021 American Chemical Society. (c) Schematic diagram for the ML-assisted design of a high-capacity organic anode. Reproduced from ref 94. Copyright 2022 American Chemical Society. (d) Development of AI-kernel for the design of high-energy-density organic cathode materials for LIBs. Reproduced with permission from ref 95. Copyright 2022 Elsevier.

throughput computational screening of ternary Li-containing fluorides (Li-M-F) for application as cathode coatings in Li-ion batteries, focusing on their phase stability, electrochemical stability, chemical stability, and ionic conductivity.[83] Ceder et al. adopted a high-throughput searching method for the identification of suitable materials for cathode coatings in all-solid-state batteries.[84] To mitigate incompatibility at the interface of SE and the electrode, a buffer layer can prevent direct contact between them. Based on the optimum structural properties of coating materials, Lu et al. screened out 3196 of such materials, which were further put under the condition of an optimized redox potential range.[73] The SoftBV package was utilized for the determination of the Li-ion conductivity of 320 materials, which finally led to 100 potential coating material for lithium−iron-phosphate cathode (Figure 2d).[73] Among these 100 promising coating materials, the authors listed 41 coating materials that perform exceptionally well compared to other materials.[73,85] Dendritic electrodeposition of lithium metal on the anode during cycling results in capacity fading and a short circuit. However, once it is initiated, it is very hard to mitigate. To address this issue, a data-driven search was carried out by Viswanathan and co-workers for the identification of promising SEs that can suppress the dendrite growth formation.[86] The ML models are trained with the structural properties of the material that do not require any first-principles calculations, which were used to screen ∼12 000 inorganic solids based on their ability to suppress the dendrite initiation in contact with the lithium metal anode. The authors trained GCNN to learn shear and bulk moduli with very high accuracy (MAEs of 0.13 and 0.10, respectively), which led them to 20 mechanical anisotropic interfaces that can suppress the dendrite growth. The relationship among composition, atomic structure, stability, and ion conductivity was unclear for the noncrystalline phases. Integration of DFT and AI-aided methods such as artificial neural network potential and genetic-algorithm sampling mapped the designing strategy of the amorphous/glassy solid electrolyte for better conductivity and stability.[87]

3.1.3. Organic Electrodes. In recent years, researchers have been looking to replace high-cost transition-metal-containing electrode materials with low-cost and highly abundant elements like C, H, and N. Hence, organic materials have attracted a lot of attention as potential electrode materials for rechargeable batteries.[88,89] However, pulling out an organic electrode with the desired redox potential from a large organic material space through a trial-and-error method is not an efficient method. Hence, researchers are utilizing ML algorithms for the screening of organic electrode materials by choosing important descriptors and suitable feature space. Since a sufficient amount of data with the desired target property is not available for organic cathode materials, efficient training and development of robust ML models have been always an issue for this kind of field study. Oaki and co-workers have carried out sparse modeling, combining ML with chemical insights to handle a small data set extracted from the literature (Figure 3a).[90] They successfully proposed BCz-PH as the potential high-voltage (4.5−4.8 V) electrode material. They showed that sparse modeling can efficiently train the ML model for the prediction of energy density, specific capacity, and reaction potential for the organic cathode materials. Although sparse modeling enables training on limited data sets, the approach may risk overfitting and might not capture the full complexity of organic electrode behavior. Immediately after describing their successful proposal of BCz-PH, Xu et al. integrated a ML workflow with an experimental flowchart for the identification of potential electrode materials based on the ML-predicted redox potential.[91] They employed the Random Forest algorithm, known for its interpretability, to predict the redox potential of organic materials using features extracted from SMILES strings. These features were subsequently analyzed using MAACS fingerprinting (Figure 3b).[92,93] While employing Random Forest with SMILES-derived features and MAACS fingerprinting is innovative, this method may oversimplify 3D molecular nuances, potentially compromising the accuracy of redox potential predictions. Screening of high-capacity anode materials through sparse modeling has been carried out to propose 5-formylsalicylic acid as a high-capacity anode material for LIBs (Figure 3c).[94] The sparse modeling approach used to propose 5-formylsalicylic acid, despite its promise, might overlook subtle nonlinear interactions due to the limited data size, warranting further experimental validation. Araujo and co-workers have developed an AI-kernel to accelerate the discovery of potential organic cathode material screening for LIBs (Figure 3d).[95] The AI-kernel has the potential to combine linear and neural model to extract knowledge from millions of organic molecules, which can predict the voltage accurately. With their proposed AI-kernel, the authors have discovered 459 high-energy-density (>1000 W h kg^{-1}) organic cathode materials. The AI-kernel's combination of linear and neural models is powerful for large-scale screening; however, the increased model complexity could obscure mechanistic insights and necessitates rigorous experimental follow-up to confirm the predicted high-energy density candidates.

Since the amount of data is a problem for organic electrode materials, applications of DNNs are stuck and the development of a suitable database of organic electrode materials with desired descriptors is needed for the full utilization of various advanced ML technologies to design potential organic electrode materials. It has been observed that sparse modeling can be very effective for the detection of suitable organic electrodes with a very small amount of data and suitable features. Simultaneously, chemical understanding of the considered features can be explored by unsupervised clustering techniques, which can reflect the importance of various functional groups present in organic molecules toward the various battery properties. Moreover, integration of chemical knowledge, data, and advanced ML techniques could be a game-changing strategy for the ultrafast design of rechargeable batteries with optimum performance.

3.2. Dual-Ion Battery. In contrast to MIBs where the working ion shuttles between two electrode materials through the electrolyte, in dual ion batteries (DIBs), the cation and anion generated from the electrolyte intercalate in anode and cathode respectively during charging.[96−99] In the past few years, researchers have been looking for post-LIB alternative energy storage devices. DIBs have emerged as potential alternative storage devices with low cost and comparable electrochemical performance.[100] Because of the amphoteric nature of the graphite, which facilitates the oxidation and reduction of electrochemical process by uptake/release of cation/anions during the charge/discharge process,[101] the dual ion batteries can be conceptualized as dual graphite batteries (DGBs) as well.

Unlike MIBs, the voltage of DIBs is influenced not only by the choice of electrode materials but also by the intercalation dynamics of both cations and anions derived from the chosen salt electrolyte. This adds a layer of complexity to the design of DIBs, where optimizing a multitude of components becomes essential. For instance, the intercalation of larger anions into the cathode typically demands higher voltages compared with smaller anions, complicating the voltage prediction and overall design strategy. To address this complexity, we have conducted a ML-based screening to identify suitable salt electrolytes for advanced DIBs.[102] This study explored a combination of 50 different salt electrolytes and 16 staging variations of graphite cathode materials paired with the corresponding anode materials, resulting in a total of 800 possible combinations for voltage determination. The study incorporated structural, elemental, and electronic features of both electrodes and electrolytes to accurately predict the voltage. The performance of various ML models was rigorously evaluated using robust cross-validation (CV) methods, including Repeated-K-Fold CV and Leave-One-Out CV (LOOCV). Among the models tested, gradient boosting-based algorithms, particularly Gradient Boosting Regressor (GBR) and Extreme Gradient Boosting Regressor (XGBR), emerged as the most suitable for voltage prediction, with test MAEs of 0.24 and 0.23 V, respectively. To determine the most generalizable model, further validation was conducted using DFT on an unseen data set. The results revealed that the XGBR model demonstrated superior generalization capabilities, effectively predicting voltage across various salt electrolyte and graphene cathode stages. Additionally, the study employed SHapley Additive Explanations (SHAP) to interpret the importance of the input features, both globally and locally. Global feature importance analysis identified the number of intercalated anions as the most critical feature, followed by the stage number and oxidation state of the cation in the salt electrolyte. This finding aligns with the understanding that anions play a dominant role in determining DIB voltage, underscoring the XGBR model's ability to efficiently map input features to the target variable. The resulting voltages further indicated that monovalent cations are associated with higher voltages compared to divalent cations. Among the tested combinations, the pairing of the sodium cation (Na^+) with the $FSICH_3$ anion exhibited the highest voltage for a DIB system consisting of a sodium anode and a graphite cathode. This study not only demonstrates the power of ML in optimizing DIB design but also provides a pathway for future advancements in battery technology through targeted electrolyte selection. The electrodes and nonaqueous electrolytes play an important role in improving the overall electrochemical properties of the DIBs. However, limited study has been done so far to design the electrode/electrolyte materials using the ML model. Das et al. has screened the active cathode materials such as different polycyclic aromatic hydrocarbons (PAHs) and different metal-

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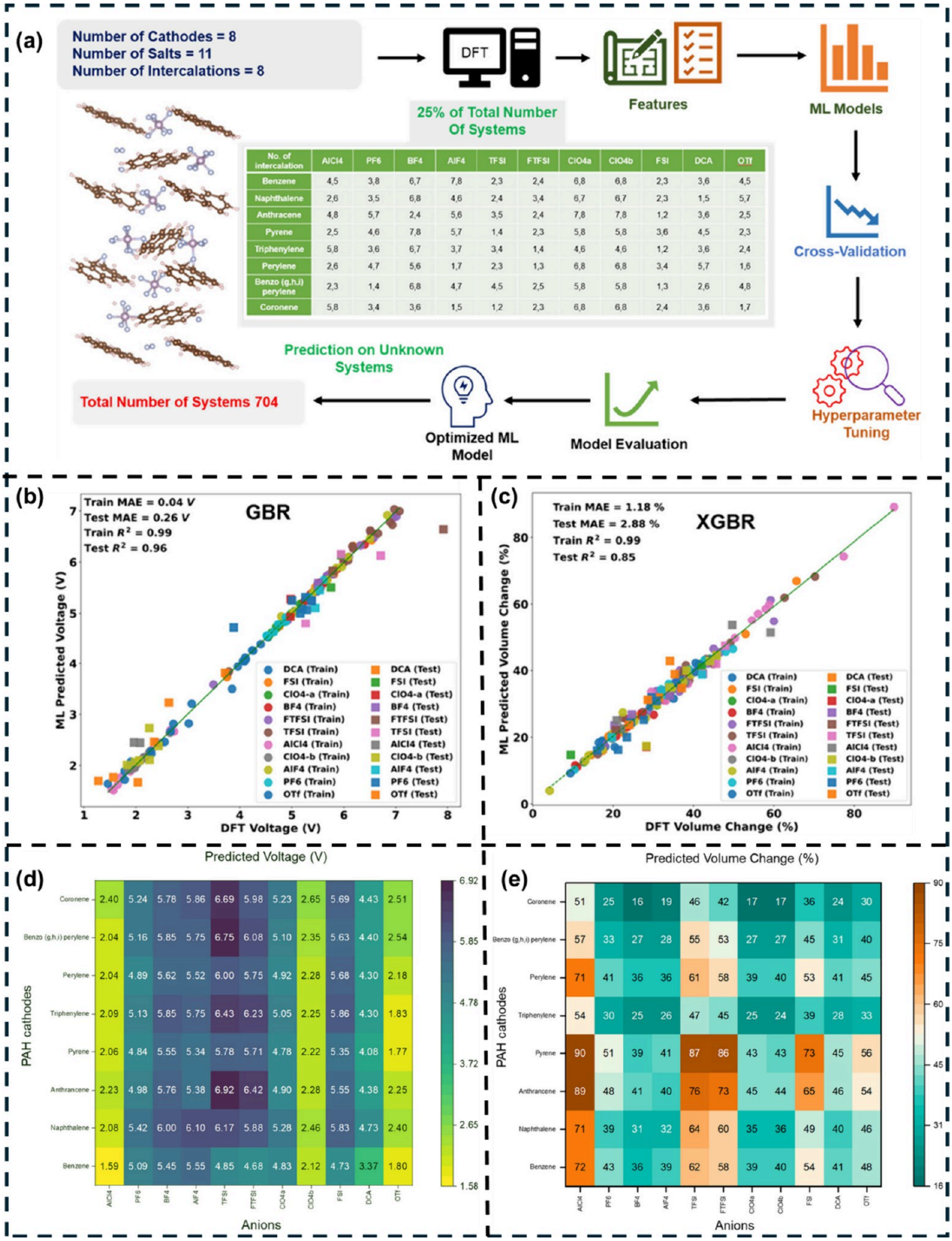

Figure 4. (a) ML workflow for predicting voltage and volume change. Scatter plot comparing DFT-calculated and ML-predicted (b) voltage and (c) volume change. Heatmap visualizing ML-predicted (d) voltage and (e) volume change for different combinations of cathodes and anions. Reproduced from ref 103. Copyright 2022 American Chemical Society.

based salt solvents as active electrolytes for the DIB study using ML models.[103] The ML workflow for the prediction of voltage and volume change is represented in Figure 4a. Both the GBR and XGBR models have been found to perform well for the prediction of voltage and volume change based on the structural and geometrical features. However, from the cross-validation result, GBR and XGBR models have been finalized for the prediction of voltage and volume change respectively for unknown systems (Figure 4b and c). The heatmaps for the average voltage and volume change of various combinations of

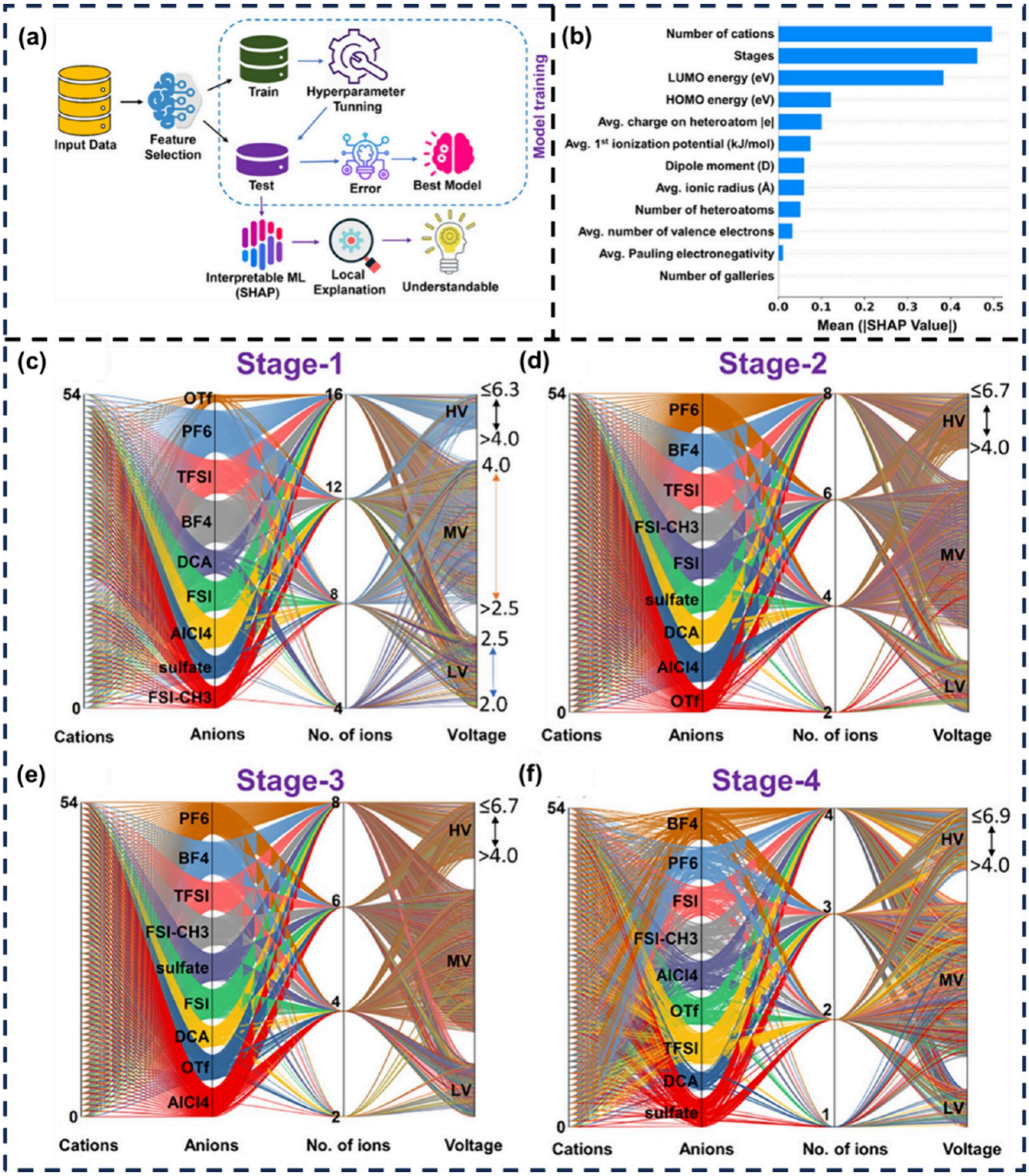

Figure 5. (a) Machine learning workflow for the prediction of voltage for different ionic liquids. (b) SHAP plot of interpretable ML models to understand global features analysis. Parallel set plots for the voltages calculated using DFT+ML for (c) stage-1, (d) stage-2, (e) stage-3, and (f) stage-4. Reproduced from ref 108. Copyright 2024 American Chemical Society.

cathodes and anions are portrayed in Figure 4d and e. The combination of $LiBF_4$ and $LiAlF_4$ salts with coronene, benzo[*g*,*h*,*i*]perylene, naphthalene, and LiFSI with coronene provides the high voltage range of 5 V. Hence, this study provides low-cost high-voltage DIBs with the most suitable pair of electrode and electrolyte combinations.

However, the use of solvent-based electrolytes in high-voltage batteries faces significant decomposition challenges. These solvents often decompose at higher voltages, leading to reduced battery performance, increased safety risks, and a shorter cycle life. Addressing these issues requires the development of different electrolytes with higher oxidative stability and implementing protective strategies such as electrolyte additives, advanced electrolyte formulations, and surface coatings for electrodes to mitigate decomposition and enhance battery longevity and safety. To prevent this decomposition of solvent-based electrolytes in DIBs, room temperature ionic liquid (RTIL) electrolytes are considered in DIB studies. Ionic liquids (ILs) are low-melting salts, and they remain in a liquid state up to 100 °C. ILs possess interesting features such as a broad electrochemical window (ECW), low vapor pressure, and high ionic conductivity, which has led to the design of another class of electrolytes with better electrochemical performance.[104,105] ECW is a critical parameter for assessing battery stability and predicting its operating voltage range. A higher ECW indicates that the electrolyte can withstand higher charging voltages, which translates to higher discharge voltages and improved battery performance. However, the calculation of ECW is challenging, and few computational methods have been developed so far, such as thermodynamic cycle approach and combined MD-DFT (AIMD-min and AIMD-sp).[106] Among these, AIMD-min has been shown to be particularly effective, yielding ECW values that align well with experimental measurements. However, despite its accuracy, AIMD-min may not be feasible for all IL electrolytes due to the high computational costs and complexity involved in accurately modeling certain IL systems. Specifically, AIMD requires intensive computational resources to capture the nuanced electronic interactions within some IL compositions, which can limit its broader applicability. Therefore, a data-driven ML model has been implemented to predict the ECW of 660 IL electrolytes in DIBs.[107] With a low MAE of 0.37 V, extra tree regression (ETR) has emerged as a suitable model for the prediction of the ECW of IL electrolytes. The ML models have been trained on both molecular and elemental features that do not need any DFT calculation and are readily available. From the ML-predicted ECW values, BF_4 and PF_6 anions containing ILs have shown to have higher ECW values that align well with the experimental report. To consider all the possible organic cation-based ILs, a data-driven ML-based model has been implemented. Using various ML models, we have successfully predicted the self-stages (cations and anions both following the same stage index) and cross stages (cations and anions following the different stages with identical number of ions) voltages of 20 780 different configurations.[108] The ML workflow that has been adopted for the prediction of voltages for various staging mechanisms is shown in Figure 5a. Among the applied ML models, the XGBR model has been found to be the most suitable model. To understand the relationship between the considered features and target variables, game theory-based SHapley Additive exPlanations (SHAP) have been considered (Figure 5b). Number of cations and stages followed by the LUMO and HOMO energy are the features that contribute the most toward the voltage prediction. By utilizing SHAP, one can roughly estimate the uncertainty in prediction by comparing the contribution of considered features with the most accurately predicted system, which can act as validation for an unknown set. We have designed 496 DGB systems, among which 69 are in the high voltage range of 4.0−7.0 V, 230 are in the moderate voltage range of 2.5− 4.0 V, and 196 belong to low-voltage range of 0.001−2.5 V.[108] The range of different voltages for various ILs considering all the different stages has been shown through a parallel plot (Figure 5c−f).

Overall, these ML-based studies serve as a guidepost to experimental researchers seeking to optimize the IL electrolytes for the advancement of the DIBs. However, other areas of the DIBs still remain unexplored, such as solid electrolyte interphase (SEI) formation and understanding the exact Fermi energy position of the electrodes with respect to their oxidative and reductive stability potentials of the IL electrolytes. Also, implementing the AI/ML model in these areas is challenging due to the absence of large data sets. Recently, the different large language model (LLM) based techniques such as natural language processing (NLP) and text mining have been implemented to extract the data from structured and unstructured sources.[109] Researchers need to consider this high potential data extraction tool to perform the ML and to understand the underlying science of the materials toward the electrochemical properties.

The solid−electrolyte interface is an integral part of a battery, which allows the migration of the working ion from the electrolyte to the electrode instead of stopping the electron flow between the electrolyte and the electrode. Designing a SEI is highly challenging, as the reaction between the electrode and the electrolyte forms a large number of organic and inorganic components, which decide the stability of the SEI. Therefore, the application of ML for design can mitigate the complexity of the understanding of the principal behind the formation of SEI. However, the lack of proper organized data related to SEI formation is also a big challenge to develop a ML model that can find the underlying pattern behind the formation of a SEI. Hence, before the application of ML, more high-throughput screening based on DFT study needs to be conducted for the establishment of a SEI database, which can further be utilized to develop a more robust and generalized ML model. Researchers have created the Lithium-Ion Battery Electrolyte (LIBE) data set, which provides accurate first-principles data on SEI species and associated reactions.[110] The data set was generated by fragmenting principal molecules, such as solvents, salts, and SEI products, and selectively recombining a subset of the fragments. These candidate molecules were analyzed using the ωB97X-V/def2-TZVPPD/SMD level of theory at various charges and spin multiplicities. The LIBE data set includes structural, thermodynamic, and vibrational information on over 17 000 unique species. This data set is expected to aid in studies of reactivity in LIBs and could be valuable for machine learning applications related to molecular and reaction properties. Wenzel and co-workers have developed an active learning workflow with a kinetic Monte Carlo model to study SEI formation in Li-ion batteries.[111] The SEI enables Li ion exchange while blocking electron transfer, impacting the battery performance. The method used a design-of-experiment approach to train a Gaussian process classification model iteratively, extending its validity and reducing the uncertainty. The challenges of stabilizing lithium metal anodes in solid-state batteries have been softened by exploring cation doping in $Li_7La_3Zr_2O_{12}$ (LLZO).[112] Using high-throughput automated reaction screening and ML, dopants such as Sc^{3+}, Ce^{3+}, Y^{3+}, and others were identified to enhance thermodynamic stability at the Li|LLZO interface, while some dopants, such as Ca^{2+} and Hf^{4+}, form passivating SEI layers to mitigate dendrite growth. The key feature influencing interface stability is the formation energy of oxides (M_xO_y), governed by M−O bond strength. The model predicts 18 unexplored LLZO systems, all validated computationally, offering a data-driven guide for experimentalists in selecting dopants for stable lithium metal anodes.

3.3. Metal−Sulfur Battery.

Despite the high theoretical capacity as well as high earth abundance of sulfur, the commercialization of metal−sulfur (M-S) batteries, specially Li−S batteries, is mainly restricted by two major issues: the shuttle effect and sluggish kinetics.[113] M-S batteries are generally composed of a metal anode and an S cathode, which typically act as a conversion type of cathode. During discharging, $Al_2Cl_7^-$ reacts with the S cathode, resulting in the formation of various metal-polysulfides for different M-S batteries. However, dissolution of metal-polysulfides into the electrolytes raises the shuttle effect.[114] Apart from the shuttle effect, the high activation energy barrier for the conversion of soluble polysulfides to

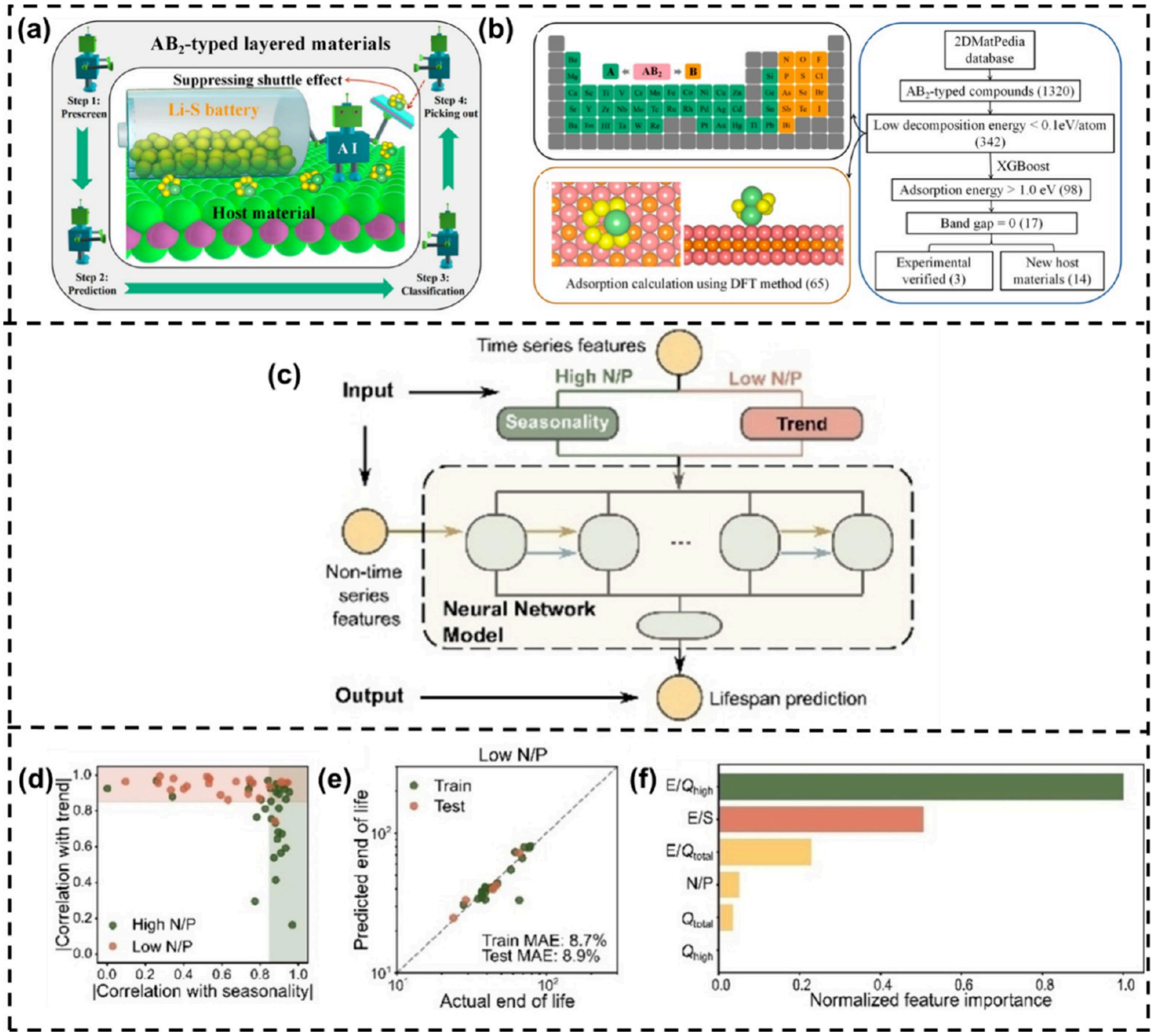

Figure 6. (a) A schematic of a machine learning method for predicting and screening suitable sulfur hosts among various materials to mitigate the shuttle effect in lithium−sulfur batteries. (b) ML workflow for screening AB_2-type materials from the 2DMatPedia database. Reproduced from ref 131. Copyright 2021 American Chemical Society. (c) Schematic illustration of the RNN-based prediction framework. (d) Absolute correlation coefficients between capacity retention and various features. (e) A parity plot illustrating the correlation between the actual and predicted end-of-life values. (f) Normalized feature importance for batteries with a low N/P ratio. Reproduced with permission from ref 132. Copyright 2022 Wiley.

insoluble polysulfides leads to capacity decay.[115−117] Catalytic conversion of these polysulfides can mitigate the shuttle effect issue through optimum binding of polysulfides as well as fast conversion into insoluble polysulfides.[118−120] Various classes of materials, such as metal-oxides, nitrides, sulfides, selenides, transition-metal-containing materials, layered double hydroxide etc., were previously reported to be utilized as supporting agents to accelerate the polysulfide conversion in Li−S battery.[121−128] However, the number of materials tested as supporting agents for M-S batteries is very limited, and testing demands innovative accelerative tools such as ML methodologies for the fast-forward discovery of a suitable catalytic agent able to inhibit the shuttle effect and sluggish kinetics. Scarcity of the data corresponding to the desired target property (adsorption energy, free energy change for polysulfide conversion) stands as a challenge for the application of ML algorithms. Though ML-assisted design of Li−S batteries has been reported to some extent, the number of reports is still very low compared to MIBs. Zhang et al. demonstrated the efficacy of transfer learning in accurately predicting the binding energies of various lithium polysulfides adsorbed on the surface of sulfur host cathode materials. Their approach accommodated different active sites and arbitrary configurations, showcasing the versatility of transfer learning in this context.[129] They have proposed that transfer learning can effectively transfer the binding energy knowledge from one material to other materials with high accuracy (MAE: 0.10 eV) if the materials belong to same class.[129] In their next investigation, they have extended their S host cathode material search space by carrying out a high-throughput screening of AB_2-type two-dimensional layered sulfur host materials, extracting 1320 materials from the 2DMatpedia database.[130] The schematic diagram of ML-assisted quick discovery of AB_2-type S host cathode materials and the corresponding workflow are shown in Figure 6a and b. The authors first refined their extracted materials based on decomposition energy and then performed a classification to classify the potential AB_2 materials for the Li−S battery. They have proposed 14 suitable AB_2 materials that can effectively bind the Li_2S_6 polysulfide, preventing dissolution in electrolytes.

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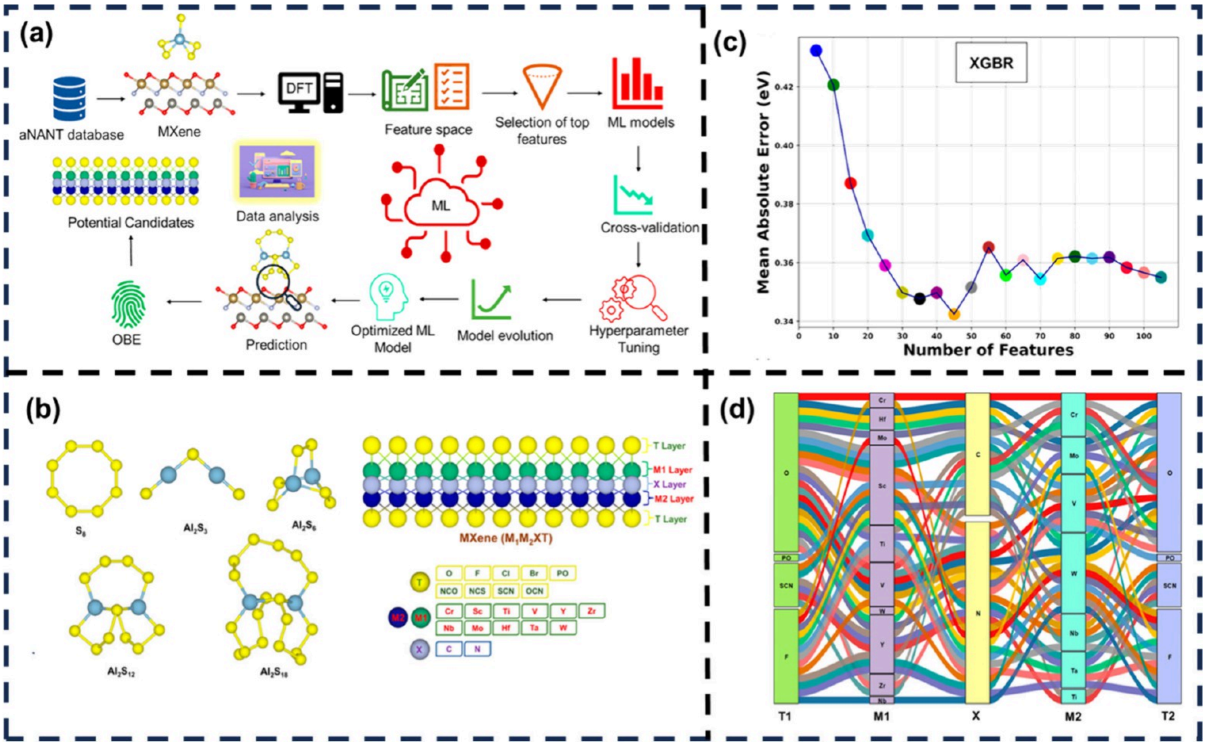

Figure 7. (a) Workflow for the prediction of the adsorption energy of Al-polysulfides using ML models. (b) Considered Al-polysulfides and MXenes. (c) Ceature optimization using the SelectKBest statistical method. (d) Alluvial plot to visualize the proposed potential S host cathode materials. Reproduced from ref 138. Copyright 2024 American Chemical Society.

Beyond AB_2 layered materials, a single-atom catalyst (SAC) can play a significant role in the improvement of Li−S battery performance. Li and co-workers have explored a large composition space of a supported SAC on nitrogen-doped carbon through ML to screen out potential S host cathode materials for Li−S batteries.[133] Liu et al. proposed an interpretable hybrid ML model composed of time series decomposition and recurrent neural network (RNN) to forecast the capacity decay in a Li−S battery.[132] The architecture of the RNN is illustrated in Figure 6c, where N/P is the capacity ratio of negative to positive electrodes. The study revealed a noteworthy distinction in the N/P ratios between two distinct groups of batteries characterized by seasonal and trend correlations (Figure 6d). It illustrates that batteries with higher N/P ratios tend to demonstrate stronger correlations with seasonal patterns, whereas those with lower N/P ratios show a greater association with trend-related features With the help of an interpretable ML model, the authors successfully elaborated the physical significance of static features such as amount of electrolyte and the loading of active material, along with the considered dynamic features (voltage curve, temperature, and internal resistance) for the prediction of capacity decay (remaining useful life and state of health) for a Li−S battery. The accuracy of the model is demonstrated in the Figure 6e through a scatter plot. In addition, they also analyzed the normalized feature importance (Figure 6f) of the considered the features, where they found feature with the highest contribution is E/Q_{high} (ratio of electrolyte volume to the discharge capacity) followed by E/S (ratio of electrolyte volume to the weight of sulfur). Thus, the hybrid interpretable ML models can convert a black box model into a gray box model. Binary descriptors (band match and lattice mismatch) have been introduced by Han et al. to represent the adsorption of Li polysulfides into a Ni-based S host cathode material, which takes care of both electronic and structural features.[134] Both these features play a crucial role in determining the polysulfide adsorption phenomena on the surface of host material, as the adsorption strength mainly controlled by the electronic coupling between adsorbed polysulfides and the active catalyst sites.[134] Apart from the Li−S battery, there are very few reports on the ML-assisted design of S host cathode materials for other M-S batteries. Currently, Na−S and Al−S batteries are starting to draw attention as alternatives to Li−S batteries. Huang and co-workers have developed a three-tier model composed of elemental-property, adsorption energy descriptor, and decomposition barrier for the ultrafast screening of graphene-supported bi-atom catalysts for Na−S batteries.[135] They found a linear correlation between the decomposition barrier and the energy difference between initial and final states of Na_2S decomposition. In recent times, two-dimensional MXene materials have been coming in handy as electrode/host materials for both MIBs and M-S batteries. There are many DFT reports on MXene-supported S host cathode materials for both Li−S and Al−S batteries.[113,136,137] Though Al−S batteries replicate similar reaction mechanisms as Li−S batteries, there is still not enough exploration and investigation for Al−S batteries. Due to the presence of various terminal groups combined with different transition metals, one can easily tune the adsorption of polysulfides on the surface of these MXene materials. However, exploration of these large complex MXene spaces through experimental and DFT needs sophisticated high-cost resources. Hence, applications of ML models to learn the descriptors are very important to also accelerate the screening process for other M-S batteries. Recently, our group has explored a large space of MXene materials through ML by choosing adsorption of Al polysulfides as the descriptor.[138] We have shown how to apply

ML models along with an optimized feature selection method and hyperparameter tuning for the prediction of the adsorption energy for a large array of MXene materials (Figure 7a). The considered Al polysulfides and structures of MXenes are shown in Figure 7b. We have predicted the adsorption energies of all the Al polysulfide intermediates with a large number of MXenes, where XGBR has been found to be the most suitable ML model with an MAE of 0.24 eV. A statistical method, namely, SelectKBest method has been adopted for the optimization of feature selection. Further, we have also analyzed the ML-predicted result through a density and distribution plot to find the effect of the terminal group present on the MXenes (Figure 7c) and based on the optimum adsorption energy criteria, potential sulfur host cathode materials have been proposed for the Al−S battery. We successfully reported the MXenes that can effectively bind the polysulfides to prevent a shuttle effect (Figure 7d). However, more intense research needs to be carried forward on the field of M-S batteries, especially for Na−S and Al−S batteries. Scarcity of data is the major issue for this field. However, a ML model trained with a small amount of data along with important descriptors can efficiently learn the adsorption phenomena as well as the kinetics of sulfur reduction. Finding out the key descriptors and their contribution is one of the main aspects that is lacking, and the utilization of interpretable ML models to explain these factors can provide much more chemical insights.

3.4. Redox Flow Battery. Redox flow batteries (RFBs) function through reversible reduction−oxidation reactions occurring between two electrolyte tanks separated by an ion-permeable membrane. During charging, one tank's redox-active species undergoes oxidation, while in the other tank the redox-active species undergoes oxidation, generating electrical energy. RFBs find applications in grid energy storage, renewable integration, and remote areas due to their scalability, long cycle life, and decoupled power and energy capacities, offering advantages over traditional battery technologies.[139] Vanadium flow batteries (VFBs) are emerging as the leading choice for large-scale energy storage, utilizing vanadium in various oxidation states (V^{3+}/V^{2+} for catholytes and VO^{2+}/VO^{2+} for anolytes).[140] Organic flow batteries (OFBs) have also garnered attention, with ongoing research focusing on designing organic redox-active species (ORASs) composed of earth-abundant elements like C, N, H, O, and S offering potential for very low-cost electrolytes, with properties such as stability, redox potential, solubility, and kinetics making OFBs a promising option for large-scale energy storage solutions.[141−143]

Optimizing parameters such as the solubility, stability, and redox potential for flow battery systems typically requires expensive experimental equipment or lengthy computational simulations. Leveraging ML algorithms offers a faster screening process by predicting these parameters based on readily available properties, facilitating efficient flow battery management. For instance, high redox potential and stability, coupled with solubility, are crucial for achieving a long cycle life and high energy density in ORASs. Establishing a database of ORASs is essential for ML-based prediction models to accurately forecast redox potential and solubility, streamlining the optimization process. Er et al. utilized a High-Throughput Virtual Screening (HTVS) approach to explore potential organic molecules as quinone derivatives.[144] Their virtual library comprised 17 core quinone structures, each adorned with 18 substituents, leading to a total of 1710 quinone/hydroquinone molecular pairs. DFT calculations were employed to assess the energy of these molecules, revealing the possibility of adjusting standard redox potentials between 0.6 and −1.5 V_{NHE}. Subsequently, 408 pairs were identified with desirable redox potential ranges. Tabor et al. expanded this research by investigating stability using DFT and semiempirical methods, analyzing over 140 000 quinone pairs for decomposition or instability mechanisms.[145] These HTVS-based studies not only investigated the potential materials but helped to increase the amount of data that can be utilized for the application of traditional and advanced ML models. The reliability of theory calculations for HTVS warrants verification, and the variability in calculation levels among researchers presents a challenge for database integration in ML applications. Hence, establishing standardized, comprehensive data samples is crucial for database sharing and ML advancements.

Aqueous solubility estimation of organic compounds has been the subject of extensive research for many years. Prediction methods generally fall into three categories: group contribution methods, regression based on molecular parameters (quantitative structure−activity relationships or QSAR), and data-driven approaches using ML algorithms such as SVM, RF, and Neural Networks (NNs), among others, to establish correlations between molecular structure and solubility.[146−151] In 2019, Sorkun et al. introduced AqSolDB, a database containing 9982 unique organic compounds, for predicting aqueous solubility.[152] In 2021, Francoeur et al. developed Sol-TranNet, a ML tool with 3393 parameters, based on the AqSolDB data set for rapid solubility prediction.[153] SolTranNet is a ML model based on a molecule attention transformer architecture that is designed to predict the aqueous solubility of drug molecules from SMILES representations. This capability is particularly relevant for the design of RFBs, where the solubility and stability of redox-active species (hydroquinone and anthraquinone) in electrolytes are critical for optimizing energy density, performance, and cycle life. Solubility dictates the concentration of redox-active species that can be dissolved in the electrolyte. Higher solubility allows for a greater amount of active material to participate in charge−discharge cycles, leading to a higher energy density. Conversely, low solubility can result in precipitation, reducing the effective capacity and risking damage to system components. For the accurate prediction of aqueous solubility, the SolTranNet model was optimized through a two-stage hyperparameter tuning process, namely, Bayesian optimization and grid search on 2D molecular representations. To mitigate the overfitting issue, a dropout layer has been introduced, leading to RMSEs of 1.459 on cross-validation and 1.711 on a test set, indicating the generalizability and stability of the ML model in the prediction of aqueous solubility. Stability is equally important, as it ensures that the redox-active species can withstand repeated charge−discharge cycles without degrading. Chemical and electrochemical stability minimize side reactions, decomposition, or precipitation, which otherwise lead to capacity fade, reduced Coulombic efficiency, and increased maintenance costs. For example, some organic molecules used in aqueous RFBs are prone to hydrolysis or oxidation, making them unsuitable for long-term operation. ML models such as SolTranNet can play a crucial role in addressing these challenges. These models can also provide insights into the structural features that govern the solubility and stability, guiding the rational design of redox-active species tailored for RFB applications. Although SolTranNet was originally developed for drug molecule solubility prediction, its architecture and performance metrics are directly relevant to redox flow batteries (RFBs). In RFBs, the solubility and stability of redox-active species are crucial

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parameters that influence the energy density and cycle life. By discussing SolTranNet, we aimed to highlight how ML models, with built-in strategies to mitigate overfitting, can be repurposed to predict these important properties, thus guiding the rational design of better-performing materials. However, the effectiveness of ML models such as SolTranNet depends on the quality and diversity of the training data. Limited representation of certain chemical spaces in data sets like AqSolDB may restrict their generalizability to unconventional RFB chemistries. Expanding these data sets and incorporating domain-specific knowledge, such as solvent−solute interactions and electrochemical behavior, can help bridge these gaps. A specific ML model named MultiDK was also devised for discovering ORASs for aqueous organic flow batteries.[154] This model combined binary descriptors (e.g., Morgan fingerprint and MACCS keys) and nonbinary descriptors (e.g., physicochemical molecular properties), resulting in significantly improved solubility prediction accuracy. Despite the development and validation of numerous solubility models, their performance on blind data sets has not consistently matched reported results, primarily due to the scarcity of high-quality experimental data and structural diversity in training data sets.[155] ML models successfully predicting the redox potential based on the structural property have also been proposed.[156−158] Doan et al. utilized the GPR model to learn the oxidation potential in homobenzylic ethers (HBEs) for nonaqueous flow batteries.[159] They curated a data set of 1400 HBEs using DFT calculations, identifying 9% with oxidation potentials falling within the desired range of 1.40−1.70 V vs NHE. They then implemented an active learning framework based on Bayesian Optimization (BO), significantly enhancing computational efficiency compared to random selection. Recently, Zhang et al. showed ML with precise theoretical calculations and experiments, revealing that indigo trisulfonate [Indigo-3(SO3H)] outperformed AQDS and predecessors in terms of solubility, capacity retention, and Coulombic efficiency.[156]

In addition to predicting properties of ORASs, ML finds application in electrode and membrane design as well as in system optimization for flow batteries (FBs). The electrode structure greatly influences the FB efficiency, with the pore structure and specific surface area playing key roles in facilitating electrochemical reactions. Wan et al. employed ML algorithms to design porous electrodes with enhanced specific surface area and hydraulic permeability for FBs.[160] They utilized a combination of stochastic reconstruction, morphological algorithms, and lattice Boltzmann methods to construct a data set comprising 2275 fibrous structures. Using algorithms such as LR, ANN, and RF, we built models to predict specific surface area and hydraulic permeability. ANN achieved the highest accuracy, with test errors of 1.91% and 11.48% for specific surface area and hydraulic permeability, respectively. Over 700 promising porous electrode materials were identified through a genetic algorithm combined with ANN. Notably, graphite-like electrodes with significantly improved specific surface area and hydraulic permeability compared to commercial ones were discovered, underscoring ML's potential in porous electrode material design for FBs. The membrane plays a pivotal role in VFBs, directly influencing their performance. Recently, researchers employed LR and ANN algorithms to forecast the effectiveness of a polybenzimidazole (PBI) porous membrane treated with various solvents. Input parameters comprised nine solvent characteristics and five experimental factors.[161] Remarkably, both the LR and ANN models achieved a mean absolute percentage error of less than 1% for voltage efficiency and energy efficiency. Experimental validation further substantiated the reliability of these models. Leveraging this predictive capability, researchers successfully identified alcohols as the most suitable solvents for modulating the porous structure of the PBI membranes.

4. MAJOR CHALLENGES

Even though researchers are actively applying and developing various ML and deep neural network models for the design of electrode materials with optimum battery performance, there are still other major challenges present that need to be addressed for complete design of battery materials. The major challenges are discussed in the following sections.

4.1. Experimental Data Prediction. Apart from theoretical property prediction, ML can also be utilized for the prediction of experimental parameters. For example, the creep rupture life is a critical parameter for evaluating the performance of Ni-based single-crystal superalloys. To accurately and efficiently predict this property, a divide-and-conquer self-adaptive (DCSA) learning method has been developed, integrating multiple material descriptors and leveraging a high-quality data set of 266 alloy samples.[162] This data set incorporates alloy composition, testing conditions, and heat treatment, along with five microstructural parameters, namely, stacking fault energy, lattice parameter, γ'-phase mole fraction, diffusion coefficient, and shear modulus, calculated using CALPHAD and structure−property relationships. The DCSA model outperforms five state-of-the-art ML models, achieving superior accuracy (RMSE = 0.38 and R^2 = 0.92). Validation on new samples shows prediction errors within an acceptable range, demonstrating its reliability. This approach offers a cost-effective and efficient alternative to experiments and holds potential for the inverse design of advanced alloys. Predicting experimental data, such as voltage profiles, capacity decay, cycle life for electrodes, and ionic conductivity or impedance for electrolytes, is a complex yet crucial challenge for the advancement of energy storage materials. While traditional first-principles methods such as DFT and molecular dynamics (MD) simulations have contributed significantly to understanding such properties, they often fall short in predicting experimental nuances due to complexities such as microstructural effects, defects, and real-world operating conditions. Most of the battery parameters' values change with time, leading to the degradation of battery performance. For example, voltage hysteresis is commonly observed in rechargeable batteries due to internal processes such as resistance and structural changes in the material during ion movement. Hysteresis leads to energy inefficiency, reducing the battery performance. Minimizing voltage hysteresis is crucial for improving the energy efficiency and lifespan, especially in electric vehicles. ML models trained on experimental data can help predict hysteresis, and NLP techniques can extract relevant data from published studies to aid in the design of materials with minimal hysteresis. Yu and co-workers introduced an innovative approach to monitoring battery degradation, focusing on estimating the entire voltage−capacity (*V*-*Q*) curve rather than relying solely on maximum capacity.[163] This method is significant, as the *V*-*Q* curve provides a more comprehensive view of battery health, encompassing phase transitions, polarization effects, and capacity loss mechanisms that cannot be captured by capacity alone. The authors propose using short-term relaxation voltages measured after brief rest periods following charge or discharge cycles as diagnostic inputs,

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offering a fast and efficient alternative to traditional methods requiring constant current cycling. Central to their approach is the use of an encoder−decoder architecture that is trained in two stages. The first stage involves unsupervised pretraining using an autoencoder, where the encoder learns key features of the relaxation voltage sequences without needing labeled data. The second stage employs transfer learning, where the pretrained encoder's weights are frozen and the decoder is fine-tuned to map relaxation voltages to corresponding *V-Q* curves. This strategy allows the model to efficiently utilize relaxation voltage data, achieving high accuracy in V-Q curve predictions with a root-mean-square error of less than 0.03 Ah. One of the most compelling aspects of this work is its ability to generate incremental capacity (IC) curves from the predicted *V-Q* data, enabling detailed insights into battery aging mechanisms. Additionally, the method is well-suited for nonconstant current charging scenarios, enhancing its practicality for diverse battery management systems, such as those in electric vehicles or grid-scale storage. By minimizing the need for extensive cycling information and leveraging quick relaxation voltage measurements, this approach could significantly advance the field of battery diagnostics, making it faster and more adaptable to dynamic charging conditions.

Advances in algorithms, modeling, and computer tech are driving lithium-ion battery design by enabling multiscale simulations, which, combined with experiments and shared data, can accelerate development and bridge key technological gaps.[164] Electrochemical impedance spectroscopy (EIS) is also a valuable tool for diagnosing lithium-ion battery health. A recent study advanced this field by predicting impedance spectra from charging voltage curves using mechanistic insights and ML.[165] The authors demonstrated that analyzing the relationships between charging curves, incremental capacity curves, and impedance spectra enables interpretable and accurate predictions with errors as low as 1.9 mΩ. This approach facilitates efficient operando impedance testing, making it practical for real-time applications such as electric vehicles. Complementing this, Iurilli et al. provided a comprehensive review of EIS applications in characterizing lithium-ion battery degradation and developing equivalent circuit models (ECMs).[166] They explored how aging test conditions influence EIS spectra and correlated spectral variations to degradation mechanisms. These insights further underscore EIS's significance in battery health diagnostics and predictive modeling. Together, these works demonstrate the synergy between experimental and ML approaches in advancing battery state-of-health evaluation.

Severson et al. explored the challenge of predicting lithium-ion battery lifetimes by analyzing early cycle discharge voltage data using ML.[167] A data set of 124 commercial lithium iron phosphate (LFP)/graphite cells, cycled under fast-charging conditions with cycle lives ranging from 150 to 2300 cycles, was used to develop predictive models. By extracting features such as the variance in changes of discharge voltage curves ($\Delta Q(V)$) between cycles, the models achieved a test error of 9.1% for regression predictions using the first 100 cycles and that of 4.9% for classification into cycle-life groups using just the first 5 cycles. The approach employs elastic net regression for a balance of accuracy and interpretability, leveraging domain knowledge while avoiding reliance on slow diagnostic cycles or specific degradation mechanisms. Voltage curve data, particularly its evolution over cycles, proved to be a rich source of predictive features, capturing subtle degradation modes before capacity fade appears. This work highlights the synergy between deliberate data generation and ML, offering a scalable and efficient route to battery diagnostics and lifecycle prediction, with implications for design, manufacturing, and optimization.

For electrolytes, ionic conductivity is one of the important metrics to determine their applicability in batteries. Hence, the development of ML models that can accurately predict the ionic conductivity of electrolytes is the need of the time to explore a larger solvent space in very short time. For instance, Shah and co-workers have trained three ML models, namely multiple linear regression, random forest, and XGBoost, on experimental ionic conductivity data extracted from the NIST ILThermo database.[168,169] This study addresses the limitations of ionic liquids (ILs) as electrolytes, particularly their high viscosity and low conductivity at lower temperatures, by developing ML models to predict their transport properties. They have considered a diverse data set of 2869 ionic conductivity values spanning 214 cations and 68 anions. The XGBoost model has been found as the most suitable model for the prediction of ionic conductivity for ionic liquids based on the features generated by the open-source cheminformatics package RDKit.[170] It has been observed that for limited data, XGBoost performs better than a Random Forest algorithm. By developing a chemistry-informed ML model, researchers accurately predicted the ionic conductivity of solid polymer electrolytes (SPEs) using data from experimental publications. The model integrates the Arrhenius equation into a message-passing neural network's readout layer, significantly enhancing predictive accuracy by accounting for temperature dependence, a key factor in ionic transport. This approach is particularly advantageous when training data are limited. The trained model was used to screen several thousand candidate SPE formulations, identifying promising options with high ionic conductivity. It also explored different anions in poly(ethylene oxide) and poly(trimethylene carbonate), offering valuable insights into the factors influencing conductivity. The integration of chemical knowledge with deep learning not only advances SPE design but also demonstrates a versatile framework for other property prediction tasks in materials science. This work highlights a pathway for accelerating the discovery of SPEs for safer, high-energy-density batteries.

Apart from these battery parameters, good models to predict experimental characterization data are crucial for bridging the gap between the material properties and their functional applications in battery design. For instance, spectroscopy data and Rietveld refined XRD peaks provide insights into the structural, electronic, and chemical properties of materials, which are vital for understanding and optimizing battery performance. Predictive models can simulate these complex data sets, accelerating the identification of promising materials, guiding synthesis, and fine-tuning their properties for specific applications. This reduces the reliance on extensive experimental trials, saving time and resources in battery development. For instance, Lee et al. have built a convolutional neural network (CNN) model trained with 1 785 405 synthetic XRD patterns that were prepared by mixing the simulated powder XRD patterns of 170 inorganic compounds from Sr−Li−Al-O quaternary compositional pool.[171] Even though the model has been trained on simulated XRD data, it can predict the experimental XRD pattern with 100% accuracy for phase identification and 86% accuracy for three-step-phase-fraction quantification. Such approaches can accelerate material screening, particularly for solid electrolytes or electrode materials, enabling the rapid design and optimization of battery components with complex compositions. Thus, ML models

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capable of learning experimental data can accelerate the field of battery design significantly in the near future.

4.2. Prediction of Crystal Structure. Crystal structure prediction (CSP) is a cornerstone of materials design, and its importance extends profoundly into the field of battery development. The ability to predict stable crystal structures for complex compositions significantly reduces the cost and time associated with experimental and computational trials. This field has witnessed remarkable advancements with the advent of ML-driven methodologies, enabling faster and more accurate predictions of complex crystal structures. These breakthroughs have implications for battery materials, especially in addressing key challenges like improving energy density, ionic conductivity, and mechanical stability. For example, Yin and co-workers have developed a robust data-driven framework that integrates high-throughput databases, such as the Open Quantum Materials Database (OQMD) and Matbench (MatB), with optimization algorithms like random searching (RAS), particle-swarm optimization (PSO), and Bayesian optimization (BO) to establish a correlation model between the crystal structure and formation enthalpies.[172] By combining these elements, their approach has successfully reduced the computational cost of structure prediction by 3 orders of magnitude compared to traditional methods. The graph network has been found to exhibit the best performance for predicting structures of 29 octet binary compounds, including some photovoltaic semiconductors such as GaAs, CdTe, and $CsPbI_3$. This flexibility and efficiency are particularly critical for batteries, where the ability to rapidly screen compositions for potential electrode or electrolyte materials can revolutionize material discovery efforts. Reliance on extensive databases and pretrained models poses a limitation when exploring various compositions outside the training space, potentially leading to gaps in the prediction accuracy for unprecedented systems.

Building on the need for efficient methods, metric learning offers another promising avenue.[173] This scalable ML approach functions as a binary classifier to distinguish whether two given compositions share similar crystal structures. Achieving an impressive accuracy of 96.4%, this method reduces the search space, making the computational prediction of stable structures more tractable. For battery design, this approach can identify structural motifs that correlate with enhanced ionic conductivity or stability, particularly in complex solid electrolytes. Yet, the simplification to binary classification might overlook finer structural nuances that are critical in applications such as cathode or anode material optimization. Recent advancements such as the machine-learning-assisted crystalline materials sampling system (MAXMAT) exemplify how integrated frameworks can streamline structure generation.[174] By leveraging PyXtal for generating potential crystal structures and M3GNET for rapidly estimating their energies, MAXMAT offers a fast and efficient approach for screening materials. This can help in battery research, where identifying stable phases capable of hosting alkali ions or facilitating fast ionic conduction is critical. For example, MAXMAT could expedite the discovery of new solid-state electrolytes with minimal interfacial resistance. The accuracy of energy estimations and the reliability of generated structures are heavily dependent on the quality of the ML potential, which may not always capture subtle electronic or ionic interactions in highly complex systems. Similarly, deep neural networks trained on crystallographic databases have been utilized to predict new compounds by analyzing templates of known structures.[175] This approach offers a pathway to discover multicomponent materials, such as high-entropy oxides or sulfides, which are being explored for advanced battery applications. Through the identification of compositions with optimal structural templates, researchers can design materials with enhanced electrochemical properties. Currently, researchers are using hybrid methods by combining traditional CSP techniques with ML to further enhance predictive capabilities. For instance, the integration of the USPEX evolutionary algorithm with ML interatomic potentials has demonstrated the ability to predict complex crystal structures, including those with more than 100 atoms in the primitive cell, with high accuracy and computational efficiency.[176] This methodology is particularly relevant for exploring polymorphs of electrode materials, where subtle structural variations can significantly impact battery performance metrics such as voltage and capacity. While this hybrid approach achieves remarkable accuracy, its computational requirements, though reduced, remain a barrier for extremely large or highly disordered systems. Recently the Δ-ML approach proposed by Wengert et al. exemplified how physics-based models and ML can synergize to address specific challenges like hydrogen bonding and many-body dispersion interactions in polymorphic systems.[177] By combining long-range physics-based models with short-range ML models, Δ-ML achieves high accuracy with a reduced computational cost. This is important for predicting structural dynamics in solid-state electrolytes or intercalation hosts under operating conditions. However, the dependency on high-quality reference data and predefined interaction models could limit its applicability to materials with unconventional bonding environments.

Generative models such as Generative Adversarial Networks (GANs) and Diffusion Models have emerged as powerful tools for creating stable, synthesizable, and realistic crystal structures. Hu and co-workers introduced a generative machine learning model called MatGAN to generate hypothetical inorganic materials with desired chemical properties.[178] It is extremely difficult to explore all possible compositions to find materials with specific properties. The authors developed MatGAN, a GAN-based model, to generate different material compositions. GANs consist of two networks that work together, a generator that creates a variety of samples and a discriminator that evaluates them. Through this interplay, the generator learns to produce increasingly realistic data, in this case, hypothetical inorganic compounds. MatGAN was trained on data from the Inorganic Crystal Structure Database (ICSD), which contains known materials and their properties. The model generates hypothetical materials not found in the ICSD data set, achieving a 92.53% uniqueness rate across 2 million generated samples. Of the generated materials, 84.5% are chemically valid. This means they are charge-neutral and have balanced electronegativity, which are essential criteria for real, stable compounds and electrode materials. Notably, the model achieves this without being explicitly programmed with these chemical rules, suggesting that it has implicitly learned composition rules during training. The introduction of CrysTens, a representation framework designed to encode crystal structures in an image-like format, captures lattice parameters, angles, space groups, and interatomic distances.[179] This encoding allows seamless integration into generative models, facilitating the generation of crystal structures while offering redundancy to mitigate noise and ensure realistic outputs. CrysTens provides a standardized approach to represent crystals that can serve as a benchmark for model performance, including the ability to produce sym-

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metrical and structurally accurate results. GANs, including their Wasserstein variants (WGANs), have demonstrated utility in generating crystal structures. However, challenges, such as training instability, mode collapse, and difficulty in maintaining crystal symmetry, often limit their effectiveness. By contrast, Diffusion Models have proven to be more robust in this context. These models consistently outperform GANs in generating symmetrical, realistic crystals, making them a promising alternative for unstructured crystal generation. Diffusion Models can propose intercalation hosts with pathways tailored for fast ion diffusion. Using classifier- or regressor-guided diffusion, materials with targeted properties such as high capacity or low overpotentials can be designed efficiently. Future developments could incorporate conditional generation into these models, enabling the direct translation of chemical formulas or natural language descriptions into crystal structures.

These advancements collectively pave the way for better battery design by providing tools to discover materials with optimized structures for specific functionalities. For example, structure prediction can identify intercalation hosts with minimal volume expansion and high ionic mobility, crucial for achieving a long cycle life and high-rate performance. Frameworks enable the discovery of solid electrolytes with high ionic conductivity and wide electrochemical stability windows. Tools like Δ-ML can be used to predict stable interfacial structures, reducing resistance and enhancing performance in solid-state batteries.

However, challenges remain. Many ML models are trained on databases that may not adequately represent the complexity of other multicomponent systems. The accuracy of predictions for systems with unconventional bonding or extreme stoichiometries is often limited. Additionally, computational methods must address not only the stability of predicted structures but also their synthesis feasibility and behavior under operating conditions. To overcome these limitations, future efforts should focus on expanding the diversity of training data sets, integrating experimental validation into prediction workflows, and developing adaptive ML models capable of handling unseen compositions. By addressing these challenges, CSP methods can truly transform battery research, enabling the rapid discovery of next-generation materials for energy storage.

4.3. Prediction of Synthesizability of Electrode Materials.

Proposing or discovering hypothetical crystal structures using advanced ML models is only the first step in the real-world design of electrode materials for various types of batteries. The hypothetically proposed materials should be synthesized experimentally to contribute toward practical applications. Synthesizability depends on factors such as thermodynamic stability, kinetic accessibility, and structural motifs, but traditional reliance on thermodynamic metrics such as decomposition energies often fails to capture the reality of metastable phases. These phases, while overlooked computationally, are experimentally accessible and are frequently essential in battery applications. For example, a graph convolutional network utilizing positive and unlabeled (PU) learning has demonstrated that structural motifs influence synthesizability beyond thermodynamic stability alone.[180] This approach achieved a true positive rate of 87.4%, validated against experimental data from the MP database, offering the potential for high-throughput screening of metastable materials that could serve as high-capacity electrodes or solid-state electrolytes. The challenge of incomplete or unlabeled data has driven the development of semisupervised learning techniques, such as the teacher-student dual neural network (TSDNN) architecture. This model significantly improves accuracy in synthesizability classification, achieving a 92.9% true positive rate and, when paired with generative models like CubicGAN, has successfully screened thousands of virtual candidates to identify stable structures validated by DFT.[181] This methodology is particularly relevant for batteries, as it narrows the search space for synthesizable materials with desirable properties, such as cubic structures with ion-conductive pathways for solid electrolytes. Thus, the combination of semisupervised learning with generative approaches offers an efficient route for designing cathode and electrolyte candidates tailored for high-performance batteries.

Understanding the experimental pathways to synthesize the predicted materials remains another critical aspect. Element-wise graph neural networks trained to predict synthesis recipes have shown top-k exact match accuracies far exceeding statistical models.[182] By prioritizing precursors and offering confidence scores, these models align computational predictions with practical experimental realization, a step essential for fabricating battery materials with complex stoichiometries. For instance, predicting synthesis pathways for layered oxide cathodes or halide-based solid electrolytes could reduce the trial-and-error typically associated with their fabrication, thus streamlining the transition from prediction to implementation. For specialized material classes, such as perovskites, which are promising for battery applications as both solid electrolytes and cathodes, ML models have demonstrated extraordinary success. A graph neural network tailored for perovskites achieved a 95.7% true positive rate for synthesizability.[183] Extending such models to other material types used in batteries, such as polyanionic frameworks or spinel oxides, could further enhance the identification of synthesizable high-capacity electrodes and fast-ion conductors. These insights into material-specific behaviors are pivotal for advancing both the efficiency and functionality of next-generation battery systems. One major obstacle in synthesizability predictions is data bias stemming from imbalanced data sets. Inconsistent training data can lead to overfitting or reduced generalizability, as evidenced by studies in anomaly detection for crystal structures.[184] Addressing this issue by curating mixed-source data sets that combine computational and experimental data has proven effective. This approach ensures more robust predictions, which is especially critical for battery applications where performance metrics, such as ionic conductivity or cycling stability, must complement synthesizability predictions. ML models that incorporate chemical principles such as charge balancing and ionicity have shown potential to link synthesizability with these performance indicators, making them valuable for identifying experimentally feasible electrode and electrolyte candidates.

Despite these advancements, challenges persist. Models often require large, labeled data sets and struggle with generalizability across diverse chemistries or structures. Additionally, the synthesizability predictions rarely account for kinetic factors or specific experimental conditions, which are crucial for certain metastable phases. Incorporating experimental data, including synthesis conditions, into training pipelines and developing ensemble models to address these complexities are promising future directions. Integrating synthesizability predictions with electrochemical performance metrics, such as energy density, conductivity, and stability, would provide a more comprehensive framework for material screening. The advancements in ML-driven synthesizability predictions, from semisupervised learning to generative and graph-based approaches, represent transformative tools for battery research. By addressing their limitations and refining their integration with performance metrics and synthesis pathways, these methodologies have the potential to accelerate the discovery of next-generation electrodes and electrolytes, ultimately bridging the gap between computational predictions and experimental realization in battery materials design.

5. SUMMARY AND PROSPECTS

Machine learning (ML) with its ultrafast screening potential can accelerate the material design exponentially compared to conventional experimental and quantum mechanical calculations. Many experimental research groups are increasingly using machine learning models to predict material properties. These predictions guide the selection of suitable electrode materials for the synthesis attempts. However, ML applications in batteries are still in their infancy, especially for dual ion batteries and metal−sulfur batteries, where the scarcity of data is the major disadvantage for the advanced application of ML models. Thus, to develop robust ML models and to increase the prediction accuracy, first, enormous efforts need to be put toward establishing accurate battery databases including both experimental and theoretical data. The experimental conditions or the methodology at which the data were generated also need to be recorded, as varying the conditions can change the specifications of battery materials performance significantly. Experiments often occur under diverse conditions, leading to significant variations in unmonitored variables. Furthermore, public data sets may be susceptible to publication bias, as negative results tend to be underrepresented despite their equal importance in training statistical models. Addressing these challenges necessitates a commitment to transparency and standardization in reporting experimental data across the scientific literature. Currently, there are only a few existing databases containing properties of metal-ion batteries, which are insufficient for the application of advanced machine learning algorithms such as GNN, CGNN, CGCNN, etc. Hence, the primary focus needs to be given to increasing the amount of MIB data in the existing database along with the development of battery databases for different types of batteries. In addition, unstructured data could be extracted from the published papers through different advanced ML methods like a natural language process (NLP) such as ChatGPT.[185−188] ChatGPT has been drawing attention very recently for the extraction of knowledge from experimental papers and utilizing the trained model for the synthesis of metal−organic-framework (MOF).[189] Utilizing the large language ML models trained on the published experimental and theoretical papers, chatbots can be created for different battery systems that then are able to bring all the important reported information in one platform.[190]

Second, the application of ML in the battery field will only be successful when the representation of materials is accurate, and this will gradually lead to the development of a robust ML model. For most of the cases, researchers introduce combinations (sum, average, deviation, and range) of various elemental properties to represent a material numerically. Consideration of elemental properties is advantageous as it does not need any quantum mechanical calculation and is readily available from the literature. However, better ways of representing the structural, electronic, and other aspects of a material should be focused on. For example, there are many materials existing in multiple stable and metastable phases where they have the same chemical formula. In these cases, the elemental properties alone cannot distinguish between two crystal structures. For such type of materials, formatting the atomic positions as feature matrix, consideration of space group, and point group can mitigate the distinguishability issue. One Hot Encoding and Label Encoding can easily convert the categorical features (space group, point group) into numerical features.[57] Currently, advanced representations such as graph representation, voxel representation, and grid representation are coming handy for the accurate and efficient representation of crystal structures.[57,191] However, these representations are mainly applicable for the DNN-based study where enough data is necessary for the learning of battery properties such as voltage, specific capacity, energy density, etc. Thus, innovative ideas are required to examine the best representation of the materials, such that they do not need complex calculations or experiments but can efficiently underscore the properties of materials.

Along with data and accurate representation of materials, advanced ML models need to be developed, which can explain the fundamental chemical principles controlling the properties of the batteries. Based on a specific battery problem, both traditional ML models and DNN models can be applicable. Mostly, where the amount of data is not very high, especially for DIBs and M-S batteries, traditional tree-based ML models (RFR, DTR, GBR, XGBR) are likely to perform better compared to complex neural network architectures. These traditional ML models are more explainable and easier to understand. Thus, the utilization of traditional ML models is always effective when the amount of data is small or moderate. However, with the increase in amount of data, neural network-based ML models have the potential to learn complex underlying chemical relationships between automated extracted features with the target variable. With a sufficient amount of data, DNN can outperform the traditional ML models for the prediction of voltage for MIBs. Further, the accuracy in voltage prediction has been increased by utilizing an advanced attention-based graph neural network where the nodes of the graph represent the atoms and the edges represent the bond between two atoms.[52] In addition to the supervised ML models, unsupervised ML models such as principal component analysis (PCA) and various clustering methods must be applied to understand the underlining pattern of the data. ML models that can handle a small amount of data are crucial in scientific battery modeling. Approaches tailored for low-data scenarios, such as data-efficient models, active sampling for data set construction, and data augmentation techniques, play pivotal roles. Incorporating uncertainty quantification, data efficiency, interpretability, and regularization enhances the robustness of ML models, ensuring their generalizability to different material classes beyond the training data distribution. Methods to quantify the distance of data points from the training set or estimate the variability in predicted labels through uncertainty quantification are valuable.

Improving interpretability by discerning feature relevance and importance aids in identifying chemically meaningful features and understanding dominant latent factors governing material properties. Global feature importance and local feature importance can shed light on the contribution of each feature toward the target variable. All the tree-based algorithms have inherent potential to analyze the importance of the considered features, known as model-dependent feature importance. However, model-independent feature importance should be given more importance, as bias on the judgment of considered

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features toward the target variable is absent. Interpretable ML models such as SHAP and Shapash are very handy for the explanation of ML predicted results based on both local and global feature analysis. These models have the potential to analyze feature importance for each individual data point, which is very useful for understanding the importance of features for the accurate prediction of battery properties.

Along with this, the inverse design of materials is currently a hot topic for the design of potential electrode materials. Generative adversarial network (GAN) is also commonly used for the acceleration of material design through the inverse design process, where the ML model can directly predict the structure of the energy storage materials. More thought should be put into how ML can be utilized for the optimization of a whole-battery design system at one time rather than only predicting a particular descriptor. Thus, development of ML models considering multiple components (cathode, anode, and electrolytes) of the battery is very important.

The development of ML methodologies and deployment of those ML models can truly accelerate the material design process. ML can be implemented efficiently as a multitasking model to solve complex chemical problems. Though time is necessary to adopt all of the different types of batteries, advanced methodologies are being reported daily for the successful implementation of ML. The integration of ML into the realm of sustainable energy holds immense potential for revolutionizing technological advancements. ML should be perceived and utilized as a complementary tool for enhancing efficiency and accelerating progress. Researchers engaged in leveraging ML for battery discovery should assess its efficacy based on the tangible advancements it facilitates in the pursuit of sustainable energy solutions. This Review mainly focused on summarizing the recent advances in ML application in battery design, and providing directions toward required improvements in various aspects of the same can motivate the research community toward realizing highly efficient energy storage systems for the ever-increasing energy consumption of the modern world.

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Notes

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